Exercises 9-1 Infinite Sequences

In problems1-6, determine whether the sequence converges or diverges, find $\lim_{n\to\infty} a_n$.

1. $a_n = \dfrac{2n^2-1}{n^2+3n-1}$

Solution: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \dfrac{2n^2-1}{n^2+3n-1} = \lim_{n\to\infty} \dfrac{2-\frac{1}{n^2}}{1+\frac{3}{n}-\frac{1}{n^2}} = 2$, so the sequence $\{a_n\}_{n=1}^{\infty}$ converges.

2. $a_n = \sin\left(\dfrac{1}{n}\right)$

Solution: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \sin\left(\dfrac{1}{n}\right) = \lim_{n\to\infty} \sin 0 = 0$, so the sequence $\{a_n\}_{n=1}^{\infty}$ converges.

3. $a_n = (-1)^n \dfrac{n}{n+2}$

Solution: Since $\lim_{k\to\infty} a_{2k} = \lim_{n\to\infty} \dfrac{2k}{2k+2} = 1$ and $\lim_{k\to\infty} a_{2k+1} = -\lim_{n\to\infty} \dfrac{2k+1}{2k+3} = -1$, we conclude that the sequence diverges.

4. $a_n = \sin(\mathrm{n}\pi)$

Solution: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \sin(n\pi) = \lim_{n\to\infty} 0 = 0$, so the sequence $\{a_n\}_{n=1}^{\infty}$ converges.

5. $a_n = \dfrac{\cos(n\pi)}{n}$

Solution: Since $-\dfrac{1}{n} \le \dfrac{\cos(n\pi)}{n} \le \dfrac{1}{n}$ and $\lim_{n\to\infty} \dfrac{1}{n} = 0$, we use squeeze theorem to conclude that $\lim_{n\to\infty} \dfrac{\cos(n\pi)}{n} = 0$.

So the sequence converges.

6. $a_n = \dfrac{(-\pi)^n}{5^n}$

Solution: Since $a_n = \dfrac{(-\pi)^n}{5^n} = \left(-\dfrac{\pi}{5}\right)^n$ and $-1 < -\dfrac{\pi}{5} < 1$, we know that $\lim_{n\to\infty} a_n = 0$, i.e. the sequence converges to 0.

7. $a_n = \dfrac{\ln n}{\sqrt{n}}$

Solution:

Since $\lim_{x\to\infty} \dfrac{\ln x}{\sqrt{x}} \overset{\frac{\infty}{\infty}}{=} \lim_{x\to\infty} \dfrac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x\to\infty} \dfrac{2}{\sqrt{x}} = 0$, we have

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \dfrac{\ln n}{\sqrt{n}} = 0$, i.e. the sequence converges to 0.

8. Write the first four terms of the sequence

$$a_1 = 2,\ a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right).$$

Then use Theorem D to show that the sequence converges.

Solution:

$a_1 = 2,\ a_2 = \dfrac{1}{2}\left(2+\dfrac{2}{2}\right) = \dfrac{3}{2},\ a_3 = \dfrac{1}{2}\left(\dfrac{3}{2}+\dfrac{4}{3}\right) = \dfrac{17}{12},$

$a_3 = \dfrac{1}{2}\left(\dfrac{17}{12}+\dfrac{24}{17}\right) = \dfrac{577}{408}.$

$a_1 = 2 > \sqrt{2}$; Since it is obvious that all a_n are positive numbers, we have

$a_{n+1} = \dfrac{1}{2}\left(a_n + \dfrac{2}{a_n}\right) \ge \sqrt{a_n \cdot \dfrac{2}{a_n}} = \sqrt{2},$

Thus $a_n \ge \sqrt{2}$ for all *n*, $\{a_n\}$ is bounded below.

Now

$a_n - a_{n+1} = a_n - \dfrac{1}{2}\left(a_n + \dfrac{2}{a_n}\right) = \dfrac{1}{2}\dfrac{a_n^{\,2}-2}{a_n} \ge 0,$

so $\{a_n\}$ is decreasing.

Then by Theorem D, we conclude that the sequence converges.

Exercises 9.2: Infinite Series

In problems 1 to 7, indicate whether the given series converges or diverges. If it converges, find its sum.

1. $\sum_{k=1}^{\infty}\left(\frac{1}{7}\right)^k$

Solution: The series is a geometric series with $a=\frac{1}{7}$ and $r=\frac{1}{7}$. So it converges and

$$\sum_{k=1}^{\infty}\left(\frac{1}{7}\right)^k=\frac{\frac{1}{7}}{1-\frac{1}{7}}=\frac{1}{6}.$$

2. $\sum_{k=0}^{\infty}\left(2\left(\frac{1}{4}\right)^k+3\left(-\frac{1}{5}\right)^k\right)$

Solution: The series $\sum_{k=0}^{\infty}\left(\frac{1}{4}\right)^k$ is a geometric series with $a=1$ and $r=\frac{1}{4}$ and the series $\sum_{k=0}^{\infty}\left(-\frac{1}{5}\right)^k$ is a geometric series with $a=1$ and $r=-\frac{1}{5}$. So $\sum_{k=0}^{\infty}\left(2\left(\frac{1}{4}\right)^k+3\left(-\frac{1}{5}\right)^k\right)$ converges and

$$\sum_{k=0}^{\infty}\left(2\left(\frac{1}{4}\right)^k+3\left(-\frac{1}{5}\right)^k\right)=2\sum_{k=0}^{\infty}\left(\frac{1}{4}\right)^k+3\sum_{k=0}^{\infty}\left(-\frac{1}{5}\right)^k$$

$$=2\cdot\frac{1}{1-\frac{1}{4}}+3\cdot\frac{1}{1-(-\frac{1}{5})}=\frac{31}{6}.$$

3. $\sum_{k=1}^{\infty}\frac{k-5}{k+2}$

Solution: Let $a_k=\frac{k-5}{k+2}$. Then

$$\lim_{k\to\infty}a_k=\lim_{k\to\infty}\frac{k-5}{k+2}=\lim_{k\to\infty}\frac{1-\frac{5}{k}}{1+\frac{2}{k}}=1\neq 0,$$ so the series diverges.

4. $\sum_{k=2}^{\infty}\left(\frac{1}{k}-\frac{1}{k-1}\right)$

Solution: Let $S_n=\sum_{k=2}^{n+1}\left(\frac{1}{k}-\frac{1}{k-1}\right)$. Then

$$S_n=(\frac{1}{2}-1)+(\frac{1}{3}-\frac{1}{2})+\cdots+(\frac{1}{n+1}-\frac{1}{n})=\frac{1}{n+1}-1,$$ so

$\lim_{n\to\infty}S_n=\lim_{n\to\infty}\left(\frac{1}{n+1}-1\right)=-1$, meaning that the series converges and $\sum_{k=1}^{\infty}\left(\frac{1}{k}-\frac{1}{k-1}\right)=-1$.

5. $\sum_{k=1}^{\infty}\frac{k!}{2^k}$

Solution: Let $a_n=\frac{n!}{2^n}$. Then $\frac{a_{n+1}}{a_n}=\frac{n+1}{2}>1$ for $n>2$. So the sequence $\{a_n\}$ is an increasing sequence for $n>2$. Then $a_n>a_2=\frac{2!}{2^2}=\frac{1}{2}>0$ for $n>2$. So the sequence $\{a_n\}$ does not converge to 0. By the nth Term Test, the series $\sum_{k=1}^{\infty}\frac{k!}{2^k}$ diverges.

6. $\sum_{k=1}^{\infty}\left(\frac{e}{\pi}\right)^{k+1}$

The series is a geometric series with $a=\left(\frac{e}{\pi}\right)^2$ and $r=\frac{e}{\pi}$, $|r|<1$. So the series converges and

$$\sum_{k=1}^{\infty}\left(\frac{e}{\pi}\right)^{k+1}=\frac{\left(\frac{e}{\pi}\right)^2}{1-\frac{e}{\pi}}=\frac{e^2}{\pi(\pi-e)}.$$

7. $\sum_{k=2}^{\infty}\left(\frac{3}{(k-1)^2}-\frac{3}{k^2}\right)$

Solution: Let $S_n=\sum_{k=2}^{n+1}\left(\frac{3}{(k-1)^2}-\frac{3}{k^2}\right)$.

$$S_n=\left(3-\frac{3}{2^2}\right)+\left(\frac{3}{2^2}-\frac{3}{3^2}\right)+\cdots+\left(\frac{3}{n^2}-\frac{3}{(n+1)^2}\right)$$

$=3-\frac{3}{(n+1)^2}$, so

$\lim_{n\to\infty}S_n=3$, meaning that the series converges, and its sum is 3.

Exercises 9.3 & 9.4: Positive Series

Use any test developed so far, to decide whether a series converges or diverges. Give a reason for your conclusion.

1. $\sum_{k=2}^{\infty}\frac{7}{4k+2}$

Solution: We will use Limit comparison test.

Let $a_k=\frac{7}{4k+2}$, $b_k=\frac{1}{k}$. Then

$$\lim_{k\to\infty}\frac{a_k}{b_k}=\lim_{k\to\infty}\frac{\frac{7}{4k+2}}{\frac{1}{k}}=\lim_{k\to\infty}\frac{7k}{4k+2}=\lim_{k\to\infty}\frac{7}{4+\frac{2}{k}}=\frac{7}{4};$$

As $0<\frac{7}{4}<\infty$ and $\sum_{k=1}^{\infty}b_k$ diverges, $\sum_{k=2}^{\infty}a_k$ diverges.

2. $\sum_{k=1}^{\infty}\frac{3}{(4+3k)^{7/6}}$

Solution: We will use the Limit comparison test.

Let $a_k=\frac{3}{(4+3k)^{7/6}}$ and $b_k=\frac{1}{k^{\frac{7}{6}}}$. Then

$$\lim_{k\to\infty}\frac{a_k}{b_k}=\lim_{k\to\infty}\frac{\frac{3}{(4+3k)^{7/6}}}{\frac{1}{k^{\frac{7}{6}}}}=\lim_{k\to\infty}\frac{3k^{\frac{7}{6}}}{(4+3k)^{7/6}}=\lim_{k\to\infty}\frac{3}{\left(\frac{4}{k}+3\right)^{\frac{7}{6}}}=\frac{3}{3^{\frac{7}{6}}}=3^{-\frac{1}{6}}.$$

As $0<3^{-\frac{1}{6}}<\infty$ and $\sum_{k=1}^{\infty}b_k$ converges, we find that $\sum_{k=1}^{\infty}a_k$ converges.

3. $\sum_{k=1}^{\infty}ke^{-3k^2}$

Solution : We will use the Ratio test.

$$\lim_{k\to\infty}\frac{a_{k+1}}{a_k}=\lim_{k\to\infty}\frac{(k+1)e^{-3(k+1)^2}}{ke^{-3k^2}}=\lim_{k\to\infty}\frac{k+1}{k}\frac{e^{3k^2}}{e^{3(k^2+2k+1)}}=1\cdot\lim_{k\to\infty}\frac{1}{e^{6k+3}}=0<1$$

The series converges.

4. $\sum_{k=1}^{\infty}\left(\frac{3}{\pi}\right)^k$

Solution: This series is a geometric series with $a=\frac{3}{\pi}$ and $r=\frac{3}{\pi}$. We have that $|r|<1$, so the series converges.

5. $\sum_{k=1}^{\infty}\left(\frac{3}{\pi}\right)^{-k}$

Solution: This series is a geometric series $\sum_{k=1}^{\infty}ar^k$ with $a=\frac{\pi}{3}$ and $r=\frac{\pi}{3}$. We have that $|r|>1$, so the series diverges.

6. $\sum_{k=1}^{\infty}\left(\frac{1}{k^2}+\frac{1}{2^k}\right)$

Solution: $\sum_{k=1}^{\infty}\frac{1}{k^2}$ is a p-series with $p=2>1$, so that series converges.

$\sum_{k=1}^{\infty}\frac{1}{2^k}$ is a geometric series with $a=\frac{1}{2}$ and $r=\frac{1}{2}$. $|r|<1$, so $\sum_{k=1}^{\infty}\frac{1}{2^k}$ converges.

Thus $\sum_{k=1}^{\infty}\left(\frac{1}{k^2}+\frac{1}{2^k}\right)$ converges.

7. $\sum_{k=1}^{\infty}\sin\frac{1}{k}$

Solution: Let $a_k=\sin\frac{1}{k}$ and $b_k=\frac{1}{k}$. Then

$\lim_{k\to\infty}\frac{a_k}{b_k}=\lim_{k\to\infty}\frac{\sin\frac{1}{k}}{\frac{1}{k}}=1$. Since $0<1<\infty$ and the harmonic series $\sum_{k=1}^{\infty}b_k$ diverges, by the Limit Comparison Test, $\sum_{k=2}^{\infty}a_k$ diverges.

8. $\sum_{k=1}^{\infty}k\sin\frac{1}{k}$

Solution: Let $u=\frac{1}{x}$. Then

$$\lim_{x\to\infty}x\sin\frac{1}{x}=\lim_{x\to\infty}\frac{\sin\frac{1}{x}}{\frac{1}{x}}=\lim_{u\to 0}\frac{\sin u}{u}=1,$$

so $\lim_{k\to\infty}k\sin\frac{1}{k}=1\neq 0$, meaning that the series diverges.

9. $\sum_{n=1}^{\infty}\frac{n}{n^2+2n+3}$

Solution: We will use the limit comparison test:

Let $a_n=\frac{n}{n^2+2n+3}$ and $b_n=\frac{1}{n}$. Then

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{n^2}{n^2+2n+3}=\lim_{n\to\infty}\frac{1}{1+\frac{2}{n}+\frac{3}{n^2}}=1.$$

As $0<1<\infty$ and $\sum_{n=1}^{\infty}b_n$ diverges, we find that $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\frac{n}{n^2+2n+3}$ diverges.

10. $\sum_{n=1}^{\infty}\frac{1}{n\sqrt{n+1}}$

Solution: We will use the limit comparison test:

Let $a_n=\frac{1}{n\sqrt{n+1}}$ and $b_n=\frac{1}{n^{3/2}}$. Then

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{n^{3/2}}{n\sqrt{n+1}}=\lim_{n\to\infty}\frac{1}{\sqrt{1+\frac{1}{n}}}=1.$$

As $0<1<\infty$ and $\sum_{n=1}^{\infty}b_n$ converges, we find that $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\frac{1}{n\sqrt{n+1}}$ converges.

11. $\sum_{n=1}^{\infty}\frac{8^n}{n!}$

Solution: We will use the ratio test: Let $a_n=\frac{8^n}{n!}$. Then

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{8^{n+1}\cdot n!}{8^n\cdot(n+1)!}=\lim_{n\to\infty}\frac{8}{n+1}=0<1,$$

so the series converges.

12. $\sum_{n=1}^{\infty}\frac{n!}{n^{100}}$

Solution: We will use the ratio test: Let $a_n=\frac{n!}{n^{100}}$. Then

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{(n+1)!\cdot n^{100}}{n!\cdot(n+1)^{100}}=\lim_{n\to\infty}\left(\frac{1}{1+\frac{1}{n}}\right)^{100}\cdot(n+1)=\infty,$$

so the series diverges.

13. $\sum_{n=1}^{\infty}\frac{(2n)^3}{n!}$

Solution: Let $a_n=\frac{(2n)^3}{n!}$. $\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{(2n+2)^3\cdot n!}{(2n)^3\cdot(n+1)!}$

$$=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^3\cdot\frac{1}{(n+1)}=0<1,$$

By the Ratio Test, the series converges.

14. $\sum_{n=1}^{\infty}\frac{n+3}{n^2\sqrt{n}}$

Solution: Let $a_n=\frac{n+3}{n^2\sqrt{n}}$ and $b_n=\frac{1}{n^{3/2}}$. As

$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{n+3}{n}=\lim_{n\to\infty}\left(1+\frac{3}{n}\right)=1$ and $\sum_{n=1}^{\infty}b_n$ converges,

we have by the limit comparison test that $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\frac{n+3}{n^2\sqrt{n}}$ converges.

15. $\sum_{n=1}^{\infty}\frac{1}{2+\sin^2 n}$

Solution: $0\le\sin^2 n\le 1$ for all n, so

$2\le 2+\sin^2 n\le 3$ and therefore $\frac{1}{2+\sin^2 n}\ge\frac{1}{3}$ for all n.

Thus, $\frac{1}{2+\sin^2 n}$ does not approach 0 as $n\to\infty$. and by the nth term test

for divergence, the series $\sum_{n=1}^{\infty}\frac{1}{2+\sin^2 n}$ diverges.

16. $\sum_{n=1}^{\infty}\frac{4+\cos n}{n^3}$

Solution: For all n, $-1\le\cos n\le 1$, meaning that $0<\frac{4+\cos n}{n^3}\le\frac{5}{n^3}$ for all n. So by the comparison test, as $\sum_{n=1}^{\infty}\frac{5}{n^3}$ converges, the series $\sum_{n=1}^{\infty}\frac{4+\cos n}{n^3}$ converges.

17. $\sum_{n=1}^{\infty}\frac{4^n+n}{n!}$

Solution: Let $a_n=\frac{4^n+n}{n!}$. Then

$$\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=\lim_{n\to\infty}\frac{(4^{n+1}+n+1)\cdot n!}{(n+1)!\cdot(4^n+n)}=\lim_{n\to\infty}\frac{1}{n+1}\left(\frac{4+\frac{n+1}{4^n}}{1+\frac{n}{4^n}}\right)=0\cdot 4=0<1,$$

so by the ratio test, the series $\sum_{n=1}^{\infty}\frac{4^n+n}{n!}$ converges.

18. $\sum_{n=1}^{\infty}\sin^2\left(\frac{1}{n}\right)$

Solution: Let $a_n=\sin^2\frac{1}{n}$ and $b_n=\frac{1}{n^2}$. Then

$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{\sin^2\frac{1}{n}}{\frac{1}{n^2}}=\lim_{n\to\infty}\left(\frac{\sin\frac{1}{n}}{\frac{1}{n}}\right)^2=1$, so by the limit comparison test, as $\sum_{n=1}^{\infty}b_n$ converges, we have that $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\sin^2\left(\frac{1}{n}\right)$ converges.

19. $\sum_{n=1}^{\infty}\sqrt{n}\left(1-\cos\left(\frac{1}{n}\right)\right)$

Solution: Let $a_n=\sqrt{n}\left(1-\cos\frac{1}{n}\right)$ and $b_n=\frac{1}{n^{3/2}}$. As

$1-\cos\frac{1}{n}\sim\frac{1}{2n^2}(n\to\infty)$, we have

$$\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}\frac{\sqrt{n}(1-\cos\frac{1}{n})}{1/n^{3/2}}=\lim_{n\to\infty}\frac{1-\cos\frac{1}{n}}{1/n^2}=\lim_{n\to\infty}\frac{1/2n^2}{1/n^2}=\frac{1}{2},$$

and $\sum_{n=1}^{\infty}b_n$ converges, we have by the limit comparison test that $\sum_{n=1}^{\infty}a_n=\sum_{n=1}^{\infty}\sqrt{n}\left(1-\cos\left(\frac{1}{n}\right)\right)$ converge

Exercise 9-5 Alternating Series, Absolute Convergence, and Conditional Convergence

In Problems 1-4, show that each alternating series convergences, and then estimate the error made by using the partial sum S_9 as an approximation to the sum S of the series.

1. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\sqrt{n}}$

Sol: $a_n=\frac{1}{\sqrt{n}}$; $\frac{1}{\sqrt{n}}>\frac{1}{\sqrt{n+1}}\Rightarrow a_n>a_{n+1}$;

$\lim_{n\to\infty}a_n=\lim_{n\to\infty}\frac{1}{\sqrt{n}}=0$, so the series converges.

The error made by using S_9 is not more than $a_{10}\approx 0.316$.

2. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{n^2+1}$

Sol: $a_n=\frac{n}{n^2+1}=\frac{1}{n+\frac{1}{n}}>\frac{1}{n+1+\frac{1}{n+1}}=a_{n+1}$,

$\lim_{n\to\infty}\frac{n}{n^2+1}=\lim_{n\to\infty}\frac{1}{n+\frac{1}{n}}=0$, the series converges.

The error made by using S_9 is not more than $a_{10}\approx 0.099$.

3. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n^2}{e^n}$

Sol: $\lim_{n\to\infty}\frac{|u_{n+1}|}{|u_n|}=\lim_{n\to\infty}\frac{(n+1)^2\cdot e^n}{n^2\cdot e^{n+1}}=\lim_{n\to\infty}\frac{1}{e}\left(\frac{1+\frac{1}{n}}{1}\right)^2=\frac{1}{e}<1$.

The series converges absolutely, hence converges.

The error made by using S_9 is not more than $a_{10}\approx 0.0045$.

4. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2^n}{n!}$

Sol: $\lim_{n\to\infty}\frac{|u_{n+1}|}{|u_n|}=\lim_{n\to\infty}\frac{2^{n+1}\cdot n!}{2^n\cdot(n+1)!}=\lim_{n\to\infty}\frac{2}{n+1}=0<1$.

The series converges absolutely, hence converges.

The error made by using S_9 is not more than $a_{10}\approx 0.00028$.

In Problems 5-9, classify each series as absolutely convergent, conditionally convergent, or divergent.

5. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{5n}$

Sol: $a_n=\frac{1}{5n}>\frac{1}{5(n+1)}=a_{n+1}$, $\lim_{n\to\infty}a_n=\lim_{n\to\infty}\frac{1}{5n}=0$, the series converges. However, the series $\sum_{n=1}^{\infty}\frac{1}{5n}$ diverges since $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges. Hence the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{5n}$ is conditionally convergent.

6. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{5n^{1.1}}$

Sol: By p-series Test $\sum_{n=1}^{\infty}\frac{1}{5n^{1.1}}$ converges. Hence $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{5n^{1.1}}$ converges absolutely.

7. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{10n+1}$

Sol: $a_n=\frac{1}{10n+1}$, $\lim_{n\to\infty}\frac{n}{10n+1}=\lim_{n\to\infty}\frac{1}{10+\frac{1}{n}}=\frac{1}{10}\neq 0$. Hence $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{10n+1}$ diverges.

8. $\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}$

Sol: $u_n=\frac{\cos n\pi}{n}=\frac{(-1)^n}{n}$, $a_n=\frac{1}{n}$, $a_{n+1}<a_n$ and $\lim_{n\to\infty}a_n=0$.

The series $\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}=\sum_{n=1}^{\infty}(-1)^n\frac{1}{n}$ converges. However, $\sum_{n=1}^{\infty}\frac{1}{n}$ diverges. Therefore, $\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}$ is conditionally convergent.

9. $\sum_{n=1}^{\infty}(-1)^n\frac{\sin n}{n\sqrt{n}}$

Sol: $|u_n|=\frac{|\sin n|}{n\sqrt{n}}<\frac{1}{n\sqrt{n}}$, $\sum_{n=1}^{\infty}\frac{1}{n\sqrt{n}}=\sum_{n=1}^{\infty}\frac{1}{n^{3/2}}$ converges. Thus $\sum_{n=1}^{\infty}(-1)^n\frac{\sin n}{n\sqrt{n}}$ converges absolutely.

Exercise 9-6 Power Series

In Problems 1-3, find the convergence set and radius of convergence for the given power series.

1. $\sum_{n=1}^{\infty}\frac{x^n}{(n-1)!}$

Sol: $\rho=\lim_{n\to\infty}\left|\frac{x^{n+1}/n!}{x^n/(n-1)!}\right|=\lim_{n\to\infty}\left|\frac{x}{n}\right|=0$, the series converges for all x. Hence, convergence set of the series is $\mathbb{R}$. and the radius of convergence is ∞

2. $\sum_{n=1}^{\infty}\frac{x^n}{n^2}$

Sol:

$$\rho=\lim_{n\to\infty}\left|\frac{x^{n+1}/(n+1)^2}{x^n/n^2}\right|=\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^2|x|=\lim_{x\to\infty}\left(\frac{1}{1+\frac{1}{n}}\right)^2|x|=|x|,$$

the series converges on $(-1,1)$.

For $x=1$, $a_n=\frac{2}{n^2}$ (p-series, $p=2$), the series converges.

For $x=-1$, $a_n=\frac{(-1)^n}{n^2}$ (alternating p-series, $p=2$), the series converges. Therefore, the series converges on $[-1,1]$ and its radius of convergence is 1.

3. $\sum_{n=1}^{\infty}(-1)^n\frac{(x-2)^n}{n}$

Sol: $\rho=\lim_{n\to\infty}\left|\frac{(x-2)^{n+1}/(n+1)}{(x-2)^n/n}\right|=\lim_{n\to\infty}\left(\frac{1}{1+\frac{1}{n}}\right)|x-2|=|x-2|$.

The series converges on $(1,3)$.

For $x=1$, $a_n=\frac{1}{n}$ (harmonic series), the series diverges.

For $x=3$, $a_n=(-1)^n\frac{1}{n}$, $\lim_{n\to\infty}\frac{1}{n}=0$ and $\frac{1}{n+1}<\frac{1}{n}$, the series converges. Hence the series converges on $(1,3]$ and its radius of convergence is 1.

In Problems 4-8, find the convergence set for the given power series. Hint: First find a formula for the *n*th term; then use the Absolute Ratio Test..

4. $\frac{x}{1\cdot2}-\frac{x^2}{2\cdot3}+\frac{x^3}{3\cdot4}-\frac{x^4}{4\cdot5}+\frac{x^5}{5\cdot6}-\cdots$

Sol: The series is equivalent to $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}x^n}{n(n+1)}$.

$\rho=\lim_{n\to\infty}\left|\frac{x^{n+1}/(n+1)(n+2)}{x^n/n(n+1)}\right|=\lim_{n\to\infty}\frac{n}{n+2}|x|=|x|$, the series converges on $(-1,1)$.

For $x=1$, the series is $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n(n+1)}$ which converges absolutely by comparison with $\sum_{n=1}^{\infty}\frac{1}{n^2}$.

For $x=-1$, the series is $\sum_{n=1}^{\infty}\frac{-1}{n(n+1)}$ which is convergent by comparison with $\sum_{n=1}^{\infty}\frac{1}{n^2}$. Hence the series converges on $[-1,1]$.

5. $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\frac{x^9}{9!}-\cdots$

Sol: The series is $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}x^{2n-1}}{(2n-1)!}$.

$\rho=\lim_{n\to\infty}\left|\frac{x^{2n+1}/(2n+1)!}{x^{2n-1}/(2n-1)!}\right|=\lim_{n\to\infty}\frac{|x^2|}{(2n+1)\cdot2n}=0$. The series converges for all x.

6. $1-x+\frac{x^2}{2}-\frac{x^3}{3}+\frac{x^4}{4}-\cdots$

Sol: The series is $1+\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{n}$.

$\rho=\lim_{n\to\infty}\left|\frac{x^{n+1}/(n+1)}{x^n/n}\right|=\lim_{n\to\infty}\left|\frac{n}{n+1}x\right|=\lim_{n\to\infty}\frac{1}{1+\frac{1}{n}}|x|=|x|$.

When $x=1$, the series is $1+\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$, so it converges.

When $x=-1$, the series is $1+\sum_{n=1}^{\infty}\frac{1}{n}$, so it diverges.

Therefore, the series converges on $(-1,1]$.

7. $\frac{x-1}{1}+\frac{(x-1)^2}{2}+\frac{(x-1)^3}{3}+\frac{(x-1)^4}{4}+\cdots$

Sol: The series is $\sum_{n=1}^{\infty}\frac{(x-1)^n}{n}$.

$\rho=\lim_{n\to\infty}\left|\frac{(x-1)^{n+1}/(n+1)}{(x-1)^n/n}\right|=\lim_{n\to\infty}\left|\frac{n}{n+1}(x-1)\right|=\lim_{n\to\infty}\left|\frac{1}{1+\frac{1}{n}}(x-1)\right|=|x-1|$

. The series converges absolutely for $|x-1|<1$, that is, $0<x<2$.

For $x=0$, the series is $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$, it converges.

For $x=2$, the series is $\sum_{n=1}^{\infty}\frac{1}{n}$, so it diverges. Hence the series converges on $[0,2)$.

8. $1+\frac{x+1}{2}+\frac{(x+1)^2}{2^2}+\frac{(x+1)^3}{2^3}+\cdots$

Sol: The series is $\sum_{n=0}^{\infty}\frac{(x+1)^n}{2^n}$.

$\rho=\lim_{n\to\infty}\left|\frac{(x+1)^{n+1}/2^{n+1}}{(x+1)^n/2^n}\right|=\frac{|x+1|}{2}$. The series converges absolutely for $\frac{|x+1|}{2}<1$, that is, $-3<x<1$.

For $x=-3$, the series is $\sum_{n=0}^{\infty}\frac{(-2)^n}{2^n}=\sum_{n=0}^{\infty}(-1)^n$, so it diverges.

For $x=1$, the series is $\sum_{n=0}^{\infty}1$ which diverges. Hence the series converges on $(-3,1)$

Exercise 9-7 Operations on Power Series
& 9-8 Taylor and Maclaurin Series

In Problems 1-5, find the power series representation for $f(x)$ and specify the radius of convergence. Each is somehow related to a geometric series.

1. $f(x)=\dfrac{1}{(1+x)^2}$

Sol: $\dfrac{1}{(1+x)^2}=-\left(\dfrac{1}{1+x}\right)'$.

$\dfrac{1}{1+x}=1-x+x^2-x^3+x^4-x^5\cdots=\sum_{n=0}^{\infty}(-1)^n x^n$, $-1<x<1$, So

$\dfrac{1}{(1+x)^2}=-(0-1+2x-3x^2+4x^3-5x^4+\cdots)$

$=1-2x+3x^2-4x^3+5x^4-\cdots=\sum_{n=1}^{\infty}(-1)^{n-1}nx^{n-1}$, $-1<x<1$

The radius of convergence is 1.

2. $f(x)=\dfrac{x^2}{1-x^4}$

Sol: $\dfrac{x^2}{1-x^4}=x^2\left(\dfrac{1}{1-x^4}\right)=x^2\sum_{n=0}^{\infty}(x^4)^n=x^2\sum_{n=0}^{\infty}x^{4n}=\sum_{n=0}^{\infty}x^{4n+2}$

$=x^2+x^6+x^{10}+x^{14}+\cdots$, $|x^4|<1\Rightarrow|x|<1$

The radius of convergence is 1.

3. $f(x)=\int_0^x\ln(1+t)dt$

Sol: Since

$\ln(1+t)=\sum_{n=1}^{\infty}(-1)^{n-1}\dfrac{t^n}{n}=t-\dfrac{t^2}{2}+\dfrac{t^3}{3}-\dfrac{t^4}{4}+\cdots$, $-1<t<1$.

$\int_0^x\ln(1+t)\,dt=\int_0^x\sum_{n=1}^{\infty}(-1)^{n-1}\dfrac{t^n}{n}dt=\sum_{n=1}^{\infty}(-1)^{n-1}\int_0^x\dfrac{t^n}{n}dt$

$=\sum_{n=1}^{\infty}(-1)^{n-1}\dfrac{x^{n+1}}{(n+1)n}=\dfrac{x^2}{2}-\dfrac{x^3}{6}+\dfrac{x^4}{12}-\dfrac{x^5}{20}+\cdots$, $-1<x<1$

The radius of convergence is 1.

4. Obtain the power series in x for $\ln[(1+x)/(1-x)]$ and specify its radius of convergence.

Hint $\ln[(1+x)/(1-x)]=\ln(1+x)-\ln(1-x)$.

Sol:

$\ln(1+x)=\sum_{n=1}^{\infty}(-1)^{n-1}\dfrac{x^n}{n}=x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+\dfrac{x^5}{5}+\cdots,-1<x\le1$

.

$\ln(1-x)=\sum_{n=1}^{\infty}(-1)^{n-1}\dfrac{(-x)^n}{n}=\sum_{n=1}^{\infty}-\dfrac{x^n}{n}=-x-\dfrac{x^2}{2}-\dfrac{x^3}{3}-\dfrac{x^4}{4}-\cdots$,

$-1<-x\le1\Rightarrow-1\le x<1$

$\ln\dfrac{1+x}{1-x}=\ln(1+x)-\ln(1-x)=2x+\dfrac{2x^3}{3}+\dfrac{2x^5}{5}+\cdots$, $-1<x<1$.

The radius of convergence is 1.

5. Find the power series in x for $f(x)=xe^{x^2}$.

Sol: $e^x=\sum_{n=0}^{\infty}\dfrac{x^n}{n!}=1+x+\dfrac{x^2}{2!}+\dfrac{x^3}{3!}+\cdots$, $x\in(-\infty,+\infty)$.

$e^{x^2}=\sum_{n=0}^{\infty}\dfrac{(x^2)^n}{n!}=\sum_{n=0}^{\infty}\dfrac{x^{2n}}{n!}$, $x\in(-\infty,+\infty)$

$xe^{x^2}=x\sum_{n=0}^{\infty}\dfrac{x^{2n}}{n!}=\sum_{n=0}^{\infty}\dfrac{x^{2n+1}}{n!}$

$=x+x^3+\dfrac{x^5}{2!}+\dfrac{x^7}{3!}+\cdots$, $x\in(-\infty,+\infty)$

6. Find the sum of $\sum_{n=1}^{\infty}nx^n$.

Solution 1: Differentiating the series for $\dfrac{1}{1-x}$ yields

$\dfrac{1}{(1-x)^2}=1+2x+3x^2+4x^3+\cdots$, multiplying this series by x

gives $\dfrac{x}{(1-x)^2}=x+2x^2+3x^3+4x^4+\cdots=\sum_{n=1}^{\infty}nx^n$. Hence

$\sum_{n=1}^{\infty}nx^n=\dfrac{x}{(1-x)^2}$ for $-1<x<1$.

Solution 2:

Let $\rho=\lim_{n\to\infty}\left|\dfrac{(n+1)x^{n+1}}{nx^n}\right|=\lim_{n\to\infty}\dfrac{n+1}{n}|x|=|x|<1\Rightarrow-1<x<1$

Let $S(x)=\sum_{n=1}^{\infty}nx^n=x\cdot\sum_{n=1}^{\infty}nx^{n-1}=x\cdot S_1(x)$

$\int_0^x S_1(x)=\int_0^x\sum_{n=1}^{\infty}nx^{n-1}dx=\sum_{n=1}^{\infty}x^n=\dfrac{x}{1-x}$, $-1<x<1$

$S_1(x)=\left(\dfrac{x}{1-x}\right)'=\dfrac{1}{(1-x)^2}$, $-1<x<1$

$S(x)=x\cdot S_1(x)=\dfrac{x}{(1-x)^2}$

7. Find the power series representation of $x/(x^2-3x+2)$.

Hint: Use partial fractions.

Sol: since $\dfrac{1}{1-x}=1+x+x^2+x^3+x^4\cdots=\sum_{n=0}^{\infty}x^n$, $-1<x<1$

$\dfrac{x}{x^2-3x+2}=\dfrac{x}{(x-1)(x-2)}=\dfrac{2}{x-2}-\dfrac{1}{x-1}=-\dfrac{1}{1-\frac{x}{2}}+\dfrac{1}{1-x}$

$=-\sum_{n=0}^{\infty}\left(\dfrac{x^n}{2}\right)+\sum_{n=0}^{\infty}x^n=\sum_{n=0}^{\infty}\left[-\left(\dfrac{x^n}{2}\right)+x^n\right]=\sum_{n=0}^{\infty}\left(1-\dfrac{1}{2^n}\right)x^n$

where $\left|\dfrac{x}{2}\right|<1$ and $|x|<1$, that is, $-1<x<1$.

8. Expand each of the following functions into a power series in $(x-a)$ at given point a and point out the interval where the expansion holds:

(1) e^x, $a=1$;

Solution $e^x=\sum_{n=0}^{\infty}\frac{x^n}{n!}$, $e^{x-1}=\sum_{n=0}^{\infty}\frac{(x-1)^n}{n!}$

$$e^x=e^{x-1+1}=e\cdot e^{x-1}=e\cdot\sum_{n=0}^{\infty}\frac{(x-1)^n}{n!}$$

$$=\sum_{n=0}^{\infty}\frac{e}{n!}(x-1)^n,\qquad x\in(-\infty,\infty).$$

(2) $\ln x$, $a=2$;

Solution: $\ln x=\ln(2+x-2)=\ln 2+\ln\left(1+\frac{x-2}{2}\right)$.

When $\frac{x-2}{2}\in(-1,1]$, that is, $x\in(0,4]$,

$$\ln\left(1+\frac{x-2}{2}\right)=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{\left(\frac{x-2}{2}\right)^n}{n}=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{(x-2)^n}{n\cdot 2^n}$$

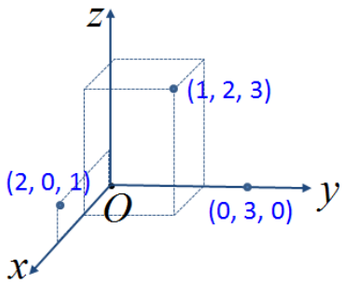
$$\ln x=\ln 2+\sum_{n=1}^{\infty}(-1)^{n-1}\frac{(x-2)^n}{n\cdot 2^n}$$

$$=\ln 2+\left[\frac{x-2}{2}-\frac{\left(\frac{x-2}{2}\right)^2}{2}+\frac{\left(\frac{x-2}{2}\right)^3}{3}-\cdots\right],\qquad x\in(0,4].$$

Exercises 11.1: Cartesian Coordinates in Three-Space

1. Plot the points whose coordinates are $(1, 2, 3)$, $(2, 0, 1)$, and $(0, 3, 0)$.

Solution:



2. Show that $(1, 1, 1)$, $(2, 2, 0)$ and $(-3,6,2)$ are vertices of a right-angled triangle. *Hint*: Only right-angled triangles satisfy the Pythagorean Theorem.

Proof: Let *A*, *B*, *C* be the points $(1, 1, 1)$, $(2, 2, 0)$ and $(-3,6,2)$ respectively. Then

$$\|\overrightarrow{AB}\| = \sqrt{(1-2)^2+(1-2)^2+(1-0)^2} = \sqrt{3},$$

$$\|\overrightarrow{AC}\| = \sqrt{(1+3)^2+(1-6)^2+(1-2)^2} = \sqrt{42} \text{ and}$$

$$\|\overrightarrow{BC}\| = \sqrt{(2+3)^2+(2-6)^2+(0-2)^2} = \sqrt{45}.$$

Since $\|\overrightarrow{AB}\|^2 + \|\overrightarrow{AC}\|^2 = 3+42 = 45 = \|\overrightarrow{BC}\|^2$, by the Pythagorean Theorem, ΔABC is a right triangle.

3. Find the distance from $(4, 5, 2)$ to (a) the $yz-$plane, (b) the point $(4,0,0)$, (c) the *x*-axis and (d) the origin.

Solution: (a) The x coordinate of a point is the directed distance from the $yz-$plane. So distance from the $yz-$plane to the point is 4.

(b) The distance between $(4, 0, 0)$ and $(4, 5, 2)$ is $\sqrt{(4-4)^2+5^2+2^2} = \sqrt{29}$.

(c) The distance from $(4, 5, 2)$ to the *y*-axis is the same to the distance from $(4, 5, 2)$ to the point on *y*-axis that is closest to the point $(4, 5, 2)$, which is $(4,0,0)$. By (b), the asked distance is $\sqrt{(4-4)^2+5^2+2^2} = \sqrt{29}$.

(d) The distance between $(0, 0, 0)$ and $(4, 5, 2)$ is $\sqrt{4^2+5^2+2^2} = \sqrt{45}$.

4. Find the equation of the sphere whose center is $(1, 2, 3)$ and that is tangent to the $xy-$plane.

Solution: Since the sphere is tangent to the xy-plane and the distance from the center to the xy-plane is 3, we conclude that the radius of the sphere is 3. Hence, the equation of the sphere is

$$(x-1)^2+(y-2)^2+(z-3)^2 = 9.$$

5. Find the equation of the sphere that has the line segment joining $(1, 1, 4)$ and $(-3, 5, -2)$ as a diameter.

Solution: The midpoint of $(1,1,4)$ and $(-3,5,-2)$ is the center of the sphere and the distance between $(1,1,4)$ and $(-3,5,-2)$ is the diameter. So the center of the sphere is $\left(\frac{1-3}{2}, \frac{1+5}{2}, \frac{4-2}{2}\right) = (-1,3,1)$ and the radius is $\frac{1}{2}\sqrt{(1+3)^2+(1-5)^2+(4+2)^2} = \sqrt{17}$. Therefore, the equation of the sphere is

6. Find the arc length of the curve given by $x = 2\cos t$, $y = 2\sin t$, $z = t$; $-\pi \le t \le \pi$.

Solution: Let L be the length of the curve. Then

$$L = \int_{-\pi}^{\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

$$= \int_{-\pi}^{\pi} \sqrt{(-2\sin t)^2 + (2\cos t)^2 + 1^2}\, dt$$

$$= \int_{-\pi}^{\pi} \sqrt{5}\, dt = 2\sqrt{5}\pi.$$

7. An object's position *P* changes so that its distance from $(1, 2, -3)$ is always twice its distance from $(1, 2, 3)$. Show that *P* is on a sphere and find its center and radius.

Proof: If the coordinates of P are (x, y, z) then because the distance from $(1, 2, -3)$ to P is twice the distance from $(1, 2, 3)$ to P, we have that

$$\sqrt{(x-1)^2+(y-2)^2+(z+3)^2} = 2\sqrt{(x-1)^2+(y-2)^2+(z-3)^2},$$

which can be rewritten as

$$(x-1)^2+(y-2)^2+(z-5)^2 = 16.$$

So P is on the sphere centered at (1, 2, 5) with radius 4.

q.e.d.

Exercises 11.2: Vectors

1. Let $\vec{u}=\langle 2,1,1\rangle$ and $\vec{v}=\langle 1,0,3\rangle$. Find the sum $\vec{u}+\vec{v}$, the difference $\vec{u}-2\vec{v}$, $\|2\vec{u}\|$ and $\|-\vec{v}\|$.

Solution: $\vec{u}+\vec{v}=\langle 3,1,4\rangle$, $\vec{u}-2\vec{v}=\langle 0,1,-5\rangle$, $\|2\vec{u}\|=2\sqrt{2^2+1^2+1^2}=2\sqrt{6}$ and $\|-\vec{v}\|=\|\vec{v}\|=\sqrt{1^2+0^2+3^2}=\sqrt{10}$.

2. Let $\vec{v}=\langle 1,3,1\rangle$. Find $\|\vec{v}\|$ and find a unit vector $\vec{u}$ with the same direction as $\vec{v}$.

Solution: $\|\vec{v}\|=\sqrt{1^2+3^2+1^2}=\sqrt{11}$. So $\vec{u}=\frac{1}{\|\vec{v}\|}\vec{v}=\frac{1}{\sqrt{11}}\langle 1,3,1\rangle$ is a unit vector and has the same direction as $\vec{v}$.

3. In the largest triangle of Figure 16, **m** is a median (a median is a line joining a vertex of a triangle to the midpoint of the opposite side). Express **m** and **n** in terms of **u** and **v**.

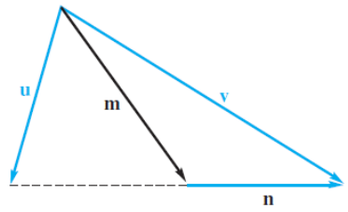


Figure 16

Solution:

$\mathbf{n}=\frac{1}{2}(\mathbf{v}-\mathbf{u})=\frac{1}{2}\mathbf{v}-\frac{1}{2}\mathbf{u}$

and $\mathbf{m}=\mathbf{v}-\mathbf{n}=\mathbf{v}-\frac{1}{2}(\mathbf{v}-\mathbf{u})=\frac{1}{2}\mathbf{v}+\frac{1}{2}\mathbf{u}$.

4. In Figure 17, $\mathbf{w}=-(\mathbf{u}+\mathbf{v})$ and $\|\mathbf{u}\|=\|\mathbf{v}\|=1$. Find $\|\mathbf{w}\|$.

Solution1 We establish the plane coordinate system as shown in the right figure. Then

$\mathbf{u}=\langle -\|\mathbf{u}\|\cos 30^\circ, \|\mathbf{u}\|\sin 30^\circ\rangle=\left\langle -\frac{\sqrt{3}}{2},\frac{1}{2}\right\rangle$,

$\mathbf{v}=\langle \|\mathbf{v}\|\cos 30^\circ, \|\mathbf{v}\|\sin 30^\circ\rangle=\left\langle \frac{\sqrt{3}}{2},\frac{1}{2}\right\rangle$.

So $\mathbf{w}=-(\mathbf{u}+\mathbf{v})=\langle 0,-1\rangle$ and $\|\mathbf{w}\|=1$.

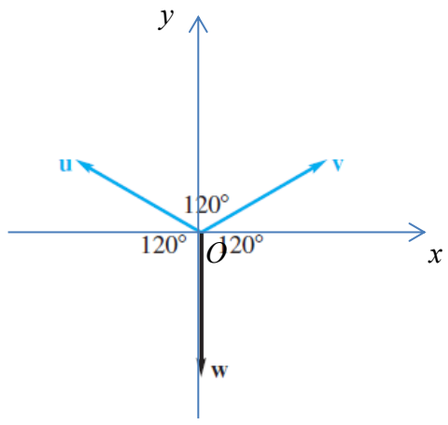


Figure 17

Solution2

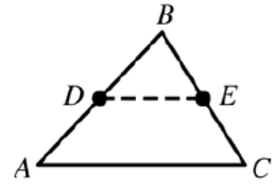
$\|\mathbf{w}\|^2=\mathbf{w}\cdot\mathbf{w}=[-(\mathbf{u}+\mathbf{v})]\cdot[-(\mathbf{u}+\mathbf{v})]=(\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v})$

$=\mathbf{u}\cdot\mathbf{u}+2\mathbf{u}\cdot\mathbf{v}+\mathbf{v}\cdot\mathbf{v}=1+2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta+1=1$

$\Rightarrow\|\mathbf{w}\|=1$.

5. Use vectors to prove that the line segment joining the midpoints of two sides of a triangle is parallel to the third side.

Proof: Given a triangle *ABC*, let *D* be the midpoint of *AB* and *E* be the midpoint of *BC*. Let $\mathbf{u}=\overrightarrow{AB}$, $\mathbf{v}=\overrightarrow{BC}$, $\mathbf{w}=\overrightarrow{AC}$ and $\mathbf{z}=\overrightarrow{DE}$.



Then

$\mathbf{w}=\mathbf{u}+\mathbf{v}$,

$\mathbf{z}=\frac{1}{2}\mathbf{u}+\frac{1}{2}\mathbf{v}=\frac{1}{2}\mathbf{w}$

Thus, *DE* is parallel to *AC*.

q.e.d.

Exercises 11.3: The Dot Product

1. Let $\vec{a}=\vec{i}+2\vec{j}+3\vec{k}$, $\vec{b}=\vec{j}+2\vec{k}$ and $\vec{c}=2\vec{i}+\vec{j}+2\vec{k}$. Find each of the following:

(a) $\vec{a}\cdot\vec{b}$ (b) $(\vec{a}+\vec{c})\cdot\vec{b}$

(c) $\vec{a}/\|\vec{a}\|$ (d) $(\vec{b}-\vec{c})\cdot\vec{a}$

(e) $\dfrac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}$ (f) $\vec{b}\cdot\vec{b}-\|\vec{b}\|^2$

Solution:

(a) $\vec{a}\cdot\vec{b}=(1)(0)+(2)(1)+(3)(2)=8$,

(b) $(\vec{a}+\vec{c})\cdot\vec{b}=(3\vec{i}+3\vec{j}+5\vec{k})\cdot(\vec{j}+2\vec{k})=13$,

(c) $\|\vec{a}\|=\sqrt{\vec{a}\cdot\vec{a}}=\sqrt{1+4+9}=\sqrt{14}$, $\vec{a}/\|\vec{a}\|=\dfrac{1}{\sqrt{14}}(\vec{i}+2\vec{j}+3\vec{k})$,

(d) $(\vec{b}-\vec{c})\cdot\vec{a}=(-2\vec{i})\cdot(\vec{i}+2\vec{j}+3\vec{k})=-2$,

(e) $\|\vec{b}\|=\sqrt{\vec{b}\cdot\vec{b}}=\sqrt{1+4}=\sqrt{5}$, $\dfrac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}=\dfrac{8}{\sqrt{14}\sqrt{5}}=\dfrac{4\sqrt{70}}{35}$,

(f) $\vec{b}\cdot\vec{b}-\|\vec{b}\|^2=\vec{b}\cdot\vec{b}-\vec{b}\cdot\vec{b}=0$.

2. Find the angle $\angle ABC$ if A has coordinates $(1,2,3)$, B has coordinates $(-4,5,6)$ and C has coordinates $(1,0,1)$.

Solution: Let $\mathbf{u}=\overrightarrow{BA}=\langle 5,-3,-3\rangle$ and $\mathbf{v}=\overrightarrow{BC}=\langle 5,-5,-5\rangle$. Then

$$\cos(\angle ABC)=\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}=\frac{(5)(5)+(-3)(-5)+(-3)(-5)}{\sqrt{25+9+9}\sqrt{25+25+25}}=\frac{11}{\sqrt{129}},$$

meaning that $\angle ABC=\arccos\dfrac{11}{\sqrt{129}}$.

3. For what numbers c and d are $\vec{u}=\langle c,2,1\rangle$ and $\vec{v}=\langle 1,-2,d\rangle$ orthogonal?

Solution: The two vectors $\vec{u}=c\vec{i}+2\vec{j}+\vec{k}$ and $\vec{v}=\vec{i}-2\vec{j}+d\vec{k}$ are orthogonal if and only if

$$\vec{u}\cdot\vec{v}=0\Leftrightarrow(c\vec{i}+2\vec{j}+\vec{k})\cdot(\vec{i}-2\vec{j}+d\vec{k})=0$$
$$\Leftrightarrow c-4+d=0.$$

So $\vec{u}$ and $\vec{v}$ are orthogonal for c and d satisfying $c+d=4$. For Example, $c=1$ and $d=3$.

4. Find $\text{proj}_{\mathbf{u}}\mathbf{v}$ if $\mathbf{u}=3\mathbf{i}+2\mathbf{j}+\mathbf{k}$ and $\mathbf{v}=2\mathbf{i}-\mathbf{k}$.

Solution:

$$\text{proj}_{\mathbf{u}}\mathbf{v}=\frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{u}\|^2}\mathbf{u}$$
$$=\frac{(2)(3)+(0)(2)+(-1)(1)}{9+4+1}(3\mathbf{i}+2\mathbf{j}+\mathbf{k})$$
$$=\frac{15}{14}3\mathbf{i}+\frac{10}{14}\mathbf{j}+\frac{5}{14}\mathbf{k}.$$

5. Given three vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ such that $\vec{u}+2\vec{v}+\vec{w}=\vec{0}$ and $\vec{u}\cdot\vec{v}=3$. If $\|\vec{u}\|=\|\vec{v}\|=2$, find $\|\vec{w}\|$.

Solution: Since $\vec{w}=-\vec{u}-2\vec{v}$,

$$\|\vec{w}\|^2=\vec{w}\cdot\vec{w}=(\vec{u}+2\vec{v})\cdot(\vec{u}+2\vec{v})$$
$$=\vec{u}\cdot\vec{u}+4\vec{u}\cdot\vec{v}+4\vec{v}\cdot\vec{v}=\|\vec{u}\|^2+4\|\vec{v}\|^2+4\vec{u}\cdot\vec{v}$$
$$=4+16+12=32.$$

So $\|\vec{w}\|=4\sqrt{2}$.

6. Find the equation of the plane passing through $(2,-3,4)$ and parallel to

(a) the $xy-$plane (b) the plane $-4x-y+2z=0$.

Solution:

(a) Planes parallel to the xy-plane can be written as $z=D$. So $z=4$ is an equation of the plane.

(b) A normal vector of the plane is $\vec{n}=\langle -4,-1,2\rangle$. An equation of the plane is

$$-4(x-2)-(y+3)+2(z-4)=0.$$

7. Find the distance from the sphere $x^2+y^2+z^2+2x+6y-8z=0$ to the plane $3x+4y+z=15$.

Solution: By completing squares, the equation of the sphere in standard form is

$$(x+1)^2+(y+3)^2+(z-4)^2=26,$$

so its center is (-1, -3, 4) and its radius is $\sqrt{26}$. The distance from the sphere to the plane is the distance from the center of the sphere to the plane minus the radius of the sphere, which is

$$\frac{|(3)(-1)+(4)(-3)+1(4)-15|}{\sqrt{3^2+4^2+1^2}}-\sqrt{26}$$
$$=\sqrt{26}-\sqrt{26}=0.$$

8. Find the equation of the sphere with center $(1,1,4)$ that is tangent to the plane $x+y=12$.

Solution: The plane with equation $x+y=12$ is parallel to the z-axis. The distance from (1, 1, 4) to the plane is the same as the distance in the xy-plane of the point (1, 1, 0) to the line $x+y-12=0$. The distance d therefore is

$$d=\frac{|1+1-12|}{\sqrt{2}}=5\sqrt{2},$$

meaning that the radius of the sphere is $5\sqrt{2}$. Therefore, the equation of the sphere is

$$(x-1)^2+(y-1)^2+(z-4)^2=50.$$

Exercises 11.4: The Cross Product

1. Let $\vec{a}=2\vec{i}+3\vec{j}-\vec{k}$, $\vec{b}=-8\vec{i}-5\vec{j}+3\vec{k}$ and $\vec{c}=-\vec{i}+2\vec{j}+\vec{k}$.
Find each of the following:

(a) $\vec{a}\times\vec{b}$ (b) $\vec{a}\times(\vec{b}+\vec{c})$

(c) $\vec{a}\cdot(\vec{b}\times\vec{c})$ (d) $\vec{a}\times(\vec{b}\times\vec{c})$

Solution:(a).

$$\vec{a}\times\vec{b}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & 3 & -1\\ -8 & -5 & 3\end{vmatrix}=(9-5)\vec{i}-(6-8)\vec{j}+(-10+24)\vec{k}$$

$$=4\vec{i}+2\vec{j}+14\vec{k}$$

(b).

$$\vec{a}\times(\vec{b}+\vec{c})=\vec{a}\times(-9\vec{i}-3\vec{j}+4\vec{k})=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & 3 & -1\\ -9 & -3 & 4\end{vmatrix}=9\vec{i}+1\vec{j}+21\vec{k}$$

(c). $$\vec{a}\cdot(\vec{b}\times\vec{c})=\vec{a}\cdot\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ -8 & -5 & 3\\ -1 & 2 & 1\end{vmatrix}=\vec{a}\cdot(-11\vec{i}+5\vec{j}-21\vec{k})=14.$$

(d). $$\vec{a}\times(\vec{b}\times\vec{c})=\vec{a}\times(-11\vec{i}+5\vec{j}-21\vec{k})=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & 3 & -1\\ -11 & 5 & -21\end{vmatrix}$$

$$=-58\vec{i}+53\vec{j}+43\vec{k}.$$

2. Find the unit vectors perpendicular to the plane determined by the three points $(1,0,1)$, $(1,-2,0)$ and $(-1,2,1)$.

Solution: The vectors $\vec{a}=\langle 1-1,0+2,1-0\rangle=\langle 0,2,1\rangle$ and $\vec{b}=\langle -1-1,2-0,1-1\rangle=\langle -2,2,0\rangle$ are both parallel to the plane. So

$$\vec{a}\times\vec{b}=(0-2)\vec{i}-(0+2)\vec{j}+(0+4)\vec{k}=-2\vec{i}-2\vec{j}+4\vec{k}$$

is perpendicular to the plane. So the unit vectors $\frac{\pm 1}{\|\vec{a}\times\vec{b}\|}\vec{a}\times\vec{b}=\pm\frac{1}{\sqrt{6}}\langle -1,-1,2\rangle$ are perpendicular to the given plane.

3. Find the area of the triangle with vertices $(3,2,1)$, $(2,4,6)$ and $(-1,2,5)$.

Solution: Let $\mathbf{a}=\langle 2-3,4-2,6-1\rangle=\langle -1,2,5\rangle$ and $\mathbf{b}=\langle -1-3,2-2,5-1\rangle=\langle -4,0,4\rangle$. The area of the triangle is half the area of the parallelogram spanned by the vectors $\mathbf{a}$ and $\mathbf{b}$. As

$$\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ -1 & 2 & 5\\ -4 & 0 & 4\end{vmatrix}=8\mathbf{i}-16\mathbf{j}+8\mathbf{k},$$

the area of the triangle is $\frac{1}{2}\sqrt{8^2+(-16)^2+8^2}=4\sqrt{6}.$

4. Find the equation of the plane through $(2,4,6)$ and perpendicular to both of the planes $3x-y+z=7$ and $-x+2y-z=-3$

Solution: The normal vector of the plane through (2,4,6) is perpendicular to the normal vectors of the other two planes. So, the vector

$$\langle 3,-1,1\rangle\times\langle -1,2,-1\rangle=\langle 1-2,-(-3+1),6-1\rangle=\langle -1,2,5\rangle$$

is a normal vector of the plane. The equation of the plane is

$$-(x-2)+2(y-4)+5(z-6)=0,$$

which can be rewritten as

$$-x+2y+5z=36.$$

5. Find the volume of the parallelepiped with edges $3\mathbf{i}-4\mathbf{j}+2\mathbf{k}$, $-\mathbf{i}+2\mathbf{j}+\mathbf{k}$, and $3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$.

Solution: The volume is

$$|\langle 3,-4,2\rangle\cdot(\langle -1,2,1\rangle\times\langle 3,-2,5\rangle)|$$
$$=|\langle 3,-4,2\rangle\cdot\langle 12,8,-4\rangle|$$
$$=|-4|=4.$$

6. Find a such that the following four points are in the same plane.

$A=(1,-1,1)$, $B=(2,1,-2)$, $C=(-1,2,2)$, $D=(1,a,a^2)$.

Solution: $\overrightarrow{AB}=\langle 1,2,-3\rangle$, $\overrightarrow{AC}=\langle -2,3,1\rangle$ and $\overrightarrow{AD}=\langle 0,a+1,a^2-1\rangle$. Since the four points are in the same plane, $\overrightarrow{AB}\times\overrightarrow{AC}$ is perpendicular to $\overrightarrow{AD}$.

Since $\overrightarrow{AB}\times\overrightarrow{AC}=\langle 2+9,-(1-6),3+4\rangle=\langle 11,5,7\rangle$,

$(\overrightarrow{AB}\times\overrightarrow{AC})\cdot\overrightarrow{AD}=5(a+1)+7(a^2-1)=(a+1)(7a-2)=0$. So $a=-1$ or $a=\frac{2}{7}$.

7 **A Pythagorean Theorem in Three-Space** As in Figure 6, let *P*, *Q*, *R*, and *O* be the vertices of a (right-angled) tetrahedron, and let *A*, *B*, *C*, and *D* be the areas of the opposite faces respectively. Show that $A^2+B^2+C^2=D^2$.

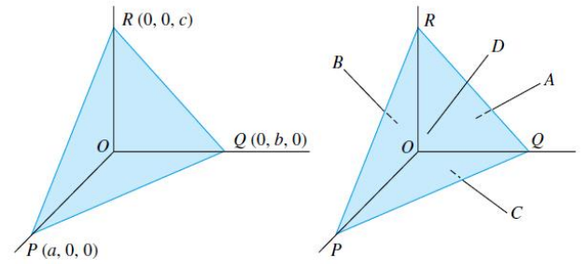


Figure 6

Solution: $\overrightarrow{PQ}=\langle -a,b,0\rangle$ and $\overrightarrow{PR}=\langle -a,0,c\rangle$, so the area of the triangle PQR is

$$\frac{1}{2}\|\overrightarrow{PQ}\times\overrightarrow{PR}\|=\frac{1}{2}\|\langle bc,ac,ab\rangle\|=\frac{1}{2}\sqrt{b^2c^2+a^2c^2+a^2b^2}.$$

So

$$D^2=\frac{1}{4}\left(b^2c^2+a^2c^2+a^2b^2\right)$$
$$=\left(\frac{1}{2}bc\right)^2+\left(\frac{1}{2}ac\right)^2+\left(\frac{1}{2}ab\right)^2=A^2+B^2+C^2.$$

q.e.d.

Exercises 11.5: Vector-Valued Functions and Curvilinear Motion

1. Find $D_t\mathbf{r}(t)$ and $D_t^2\mathbf{r}(t)$ if

(1) $\mathbf{r}(t)=(3t-4)^3\,\mathbf{i}+e^{t^2}\mathbf{j}+\mathbf{k}$

Solution:

$$D_t\mathbf{r}(t)=9(3t-4)^2\mathbf{i}+2te^{t^2}\mathbf{j}+0\mathbf{k}=9(3t-4)^2\mathbf{i}+2te^{t^2}\mathbf{j},$$

$$D_t^2\mathbf{r}(t)=54(3t-4)\mathbf{i}+2(2t^2+1)e^{t^2}\mathbf{j}.$$

(2) $\mathbf{r}(t)=\sin^2 t\mathbf{i}+\cos 3t\mathbf{j}+t^2\mathbf{k}$

Solution:

$$D_t\mathbf{r}(t)=(\sin 2t)\mathbf{i}-3(\sin 3t)\mathbf{j}+2t\mathbf{k},$$

$$D_t^2\mathbf{r}(t)=2(\cos 2t)\mathbf{i}-9(\cos 3t)\mathbf{j}+2\mathbf{k}.$$

2. If $\mathbf{r}(t)=\sqrt{t-1}\mathbf{i}+\ln(2t^2)\mathbf{j}$ and $h(t)=e^{-3t}$, find $D_t[h(t)\mathbf{r}(t)]$.

Solution:

$$h(t)\mathbf{r}(t)=e^{-3t}\sqrt{t-1}\,\mathbf{i}+e^{-3t}\ln(2t^2)\mathbf{j},\ \text{so}$$

$$D_t[h(t)\mathbf{r}(t)]=\left(-3e^{-3t}\sqrt{t-1}+\frac{1}{2}e^{-3t}(t-1)^{-\frac{1}{2}}\right)\mathbf{i}$$

$$+\left(-3e^{-3t}\ln(2t^2)+e^{-3t}\frac{2}{t}\right)\mathbf{j}$$

$$=-\frac{e^{-3t}}{2}\left(\frac{6t-7}{\sqrt{t-1}}\right)\mathbf{i}+e^{-3t}\left(\frac{2}{t}-3\ln(2t^2)\right)\mathbf{j}.$$

3. Find the velocity $\mathbf{v}$, the acceleration $\mathbf{a}$ and the speed $\frac{ds}{dt}$ at the indicated time $t=t_1$.

(1) $\mathbf{r}(t)=\cos t\,\mathbf{i}+\sin t\,\mathbf{j}+t\,\mathbf{k};\ t_1=\pi$

Solution:

$$\mathbf{v}(t)=\mathbf{r}'(t)=-\sin t\,\mathbf{i}+\cos t\,\mathbf{j}+\mathbf{k},$$

$$\mathbf{a}(t)=\mathbf{v}'(t)=-\cos t\,\mathbf{i}-\sin t\,\mathbf{j},$$

$$\text{so } \mathbf{v}(\pi)=-\mathbf{j}+\mathbf{k},\ \mathbf{a}(\pi)=\mathbf{i}$$

$$\left.\frac{ds}{dt}\right|_{t=\pi}=\|\mathbf{v}(\pi)\|=\sqrt{1+1}=\sqrt{2}\approx 1.414.$$

(2). $\mathbf{r}(t)=\ln t\,\mathbf{i}+\ln t^2\mathbf{j}+\ln t^3\mathbf{k};\ t_1=2$

Solution:

$$\mathbf{v}(t)=\mathbf{r}'(t)=\frac{1}{t}\mathbf{i}+\frac{2}{t}\mathbf{j}+\frac{3}{t}\mathbf{k},$$

$$\mathbf{a}(t)=\mathbf{v}'(t)=-\frac{1}{t^2}\mathbf{i}-\frac{2}{t^2}\mathbf{j}-\frac{3}{t^2}\mathbf{k},$$

$$\mathbf{v}(2)=\frac{1}{2}\mathbf{i}+\mathbf{j}+\frac{3}{2}\mathbf{k},$$

$$\mathbf{a}(2)=-\frac{1}{4}\mathbf{i}-\frac{1}{2}\mathbf{j}-\frac{3}{4}\mathbf{k},$$

$$\left.\frac{ds}{dt}\right|_{t=2}=\|\mathbf{v}(2)\|=\sqrt{\frac{1}{4}+1+\frac{9}{4}}=\frac{\sqrt{14}}{2}\approx 1.871.$$

4. Prove that $\|\mathbf{r}(t)\|$ is a constant if and only if $\mathbf{r}(t)\cdot\mathbf{r}'(t)=0$.

Proof: If $\|\mathbf{r}(t)\|=C$, then $\|\mathbf{r}(t)\|^2=\mathbf{r}(t)\cdot\mathbf{r}(t)=C$. Differentiating both sides, we get

$$\mathbf{r}'(t)\cdot\mathbf{r}(t)+\mathbf{r}(t)\cdot\mathbf{r}'(t)=0,$$

giving,

$$\mathbf{r}(t)\cdot\mathbf{r}'(t)=0.$$

Conversely, if $\mathbf{r}(t)\cdot\mathbf{r}'(t)=0$ we have $2\mathbf{r}(t)\cdot\mathbf{r}'(t)=0$. As this means that

$$D_t[\mathbf{r}(t)\cdot\mathbf{r}(t)]=2\mathbf{r}(t)\cdot\mathbf{r}'(t)=0,$$

we have that $\mathbf{r}(t)\cdot\mathbf{r}(t)=\|\mathbf{r}(t)\|^2$ is a constant. So $\|\mathbf{r}(t)\|$ is a constant.

q.e.d

Exercises 11.6: Lines and Tangent Lines in Three-Space

1. Write both the parametric equations and the symmetric equations for the line through the given point $(2,-1,5)$ and parallel to the given vector $\langle 1,-3,1\rangle$.

Solution: The parametric equations of the line are

$$x=2+t,\ y=-1-3t,\ z=5+t.$$

Its symmetric equations are

$$\frac{x-2}{1}=\frac{y+1}{-3}=\frac{z-5}{1}.$$

2. Find the symmetric equations of the line tangent to the curve with equation

$$\mathbf{r}(t)=2\cos t\mathbf{i}+6\sin t\mathbf{j}+t\mathbf{k}$$

at $t=\pi/3$.

Solution: $\mathbf{r}\left(\frac{\pi}{3}\right)=\mathbf{i}+3\sqrt{3}\mathbf{j}+\frac{\pi}{3}\mathbf{k}$, so $\left(1,3\sqrt{3},\frac{\pi}{3}\right)$ lies on the tangent line. $\mathbf{r}'(t)=-2\sin t\mathbf{i}+6\cos t\mathbf{j}+\mathbf{k}$, so $\mathbf{r}'\left(\frac{\pi}{3}\right)=-\sqrt{3}\mathbf{i}+3\mathbf{j}+\mathbf{k}$ is a direction vector of the tangent line at $t=\frac{\pi}{3}$. The symmetric equations of the line tangent to the curve at $t=\frac{\pi}{3}$ are therefore

$$\frac{x-1}{-\sqrt{3}}=\frac{y-3\sqrt{3}}{3}=\frac{z-\pi/3}{1}.$$

3. Find the equation of the plane perpendicular to the curve $x=3t,\ y=2t^2,\ z=t^5$ at $t=-1$.

Solution: When t = -1, we get the point (-3, 2, -1) that lies in the plane. Let $\mathbf{r}(t)=3t\mathbf{i}+2t^2\mathbf{j}+t^5\mathbf{k}$. Then $\mathbf{r}'(t)=3\mathbf{i}+4t\mathbf{j}+5t^4\mathbf{k}$, so $\mathbf{r}'(-1)=3\mathbf{i}-4\mathbf{j}+5\mathbf{k}$ is tangent to the curve at $t=-1$, so it is normal to the plane. The equation of the plane thus is

$$3(x+3)-4(y-2)+5(z+1)=0,$$

or equivalently

$$3x-4y+5z=-22.$$

4. Consider the curve $\mathbf{r}(t)=2t\mathbf{i}+t^2\mathbf{j}+\left(1-t^2\right)\mathbf{k}$

(a) Show that this curve lies in a plane and find the equation of this plane.

(b) Where does the tangent line at $t=2$ intersect the $xy-$ plane?

Solution: (a). **Method I**: Let $x(t)=2t,\ y(t)=t^2,\ z(t)=1-t^2$.

Then for any t, $y(t)+z(t)=1$,

That implies that the coordinates $(x(t),\ y(t),\ z(t))$ of any point on the curve satisfy the equation $y+z=1$, which is an equation for a plane.

Method II: We first find three points in the curve.

Let $t=0$, we have $P(0,0,1)$;

Let $t=1$, we have $Q(2,1,0)$;

Let $t=-1$, we have $R(-2,1,0)$.

Next, we find the equation of the plane passing through P,Q,R.

$\overrightarrow{RP}=\langle 2,-1,1\rangle;\overrightarrow{RQ}=\langle 4,0,0\rangle;$

$$\overrightarrow{RP}\times\overrightarrow{RQ}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & -1 & 1\\ 4 & 0 & 0\end{vmatrix}=4\vec{j}+4\vec{k}=4\langle 0,1,1\rangle.$$

The normal vector of the plane can be taken to be $\vec{n}=\langle 0,1,1\rangle$.

The equation of the plane passing through P,Q,R is $0(x-0)+1(y-0)+1(z-1)=0$, i.e. $y+z=1$.

Let $M(2t,t^2,1-t^2)$ be any point in the curve, which satisfies $t^2+1-t^2=1$.

Therefore, the curve lies in the plane $y+z=1$.

(b)

$\mathbf{r}(2)=4\mathbf{i}+4\mathbf{j}-3\mathbf{k},\mathbf{r}'(t)=2\mathbf{i}+2t\mathbf{j}-2t\mathbf{k}$ and $\mathbf{r}'(2)=2\mathbf{i}+4\mathbf{j}-4\mathbf{k}$,

so the parametric equations of the tangent line at $t=2$ are

$x=4+2t,\ y=4+4t$ and $z=-3-4t$.

The line intersects the xy-plane when $z=0$, that is,

$t=-\frac{3}{4}$. Inserting $t=-\frac{3}{4}$ into the parametric equations

gives $x=\frac{5}{2},\ y=1$ and $z=0$, meaning that the tangent line at

$t=2$ intersects the $xy-$plane at $\left(\frac{5}{2},1,0\right)$.

5. Find the distance between the two lines

$x=2-t,\ y=3+3t,\ z=2t$ and $x=-1+t, y=2-3t, z=-1-2t$.

Hint: The two lines are parallel.

Solution: When t = 0, we get the point A(2, 3, 0) is on the first line and the plane P: $-(x-2)+3(y-3)+2z=0$ is perpendicular to the two lines. The second line intersects the plane P at B$\left(\frac{-8}{7},\frac{17}{7},\frac{-5}{7}\right)$. The line segment AB is perpendicular to the two lines. So, the distance between A and B is the distance between the two parallel lines. The distance is $\sqrt{\left(2+\frac{8}{7}\right)^2+\left(3-\frac{17}{7}\right)^2+\left(\frac{5}{7}\right)^2}=\frac{5}{7}\sqrt{21}$.

6. Find the distance between the skew (nonintersecting and nonparallel) lines with parametric equations $x=2-t,\ y=3+4t,\ z=2t$ and $x=-1+t$, $y=2,\ z=-1+2t$ by taking the following steps:

(a) Note by letting $t=0$ that $(2,\ 3,\ 0)$ lies on the first line.

(b) Find the equation of the plane π through $(2,\ 3,\ 0)$ parallel to both given lines (i.e. the normal of π is perpendicular to both lines).

(c) Find a point Q on the second line.

(d) Find the distance from Q to the plane π.

Solution: (a). Let $t=0$. Then the parametric equations of the first line give

$$x=2-t=2,\ y=3+4t=3 \text{ and } z=2t=0,$$

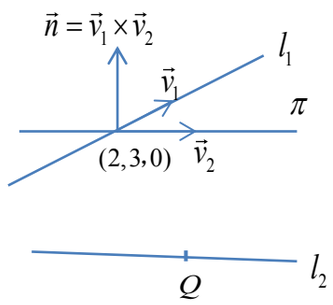
so $(2,3,0)$ lies on the first line.

(b). As the normal of the plane π, we choose

$$\langle -1,4,2\rangle\times\langle 1,0,2\rangle=\langle 8,4,-4\rangle.$$

So the equation of π is $8(x-2)+4(y-3)-4(z-0)=0$, which can be rewritten as $2x+y-z=7$.

(c). Let $t=0$. Then the parametric equations of the second line give $x=-1,\ y=2$ and $z=-1$. We choose the point Q to be $(-1,2,-1)$.



(d). Using the formula from Example 10 of Section 11.3 of our books, we find that the distance from Q to π is

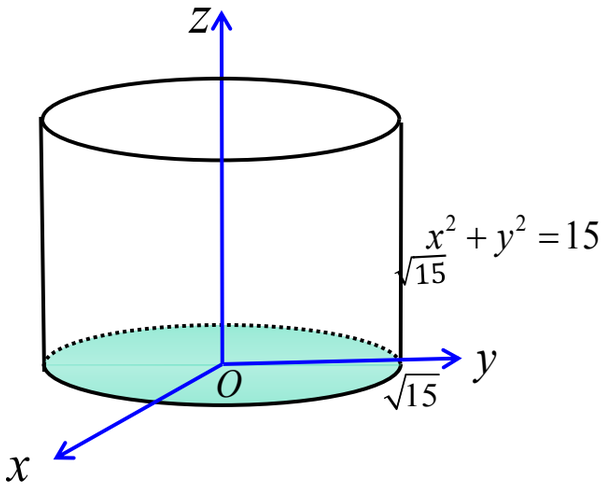
$$\frac{|2(-1)+2-(-1)-7|}{\sqrt{4+1+1}}=\sqrt{6}.$$

Exercises 11.8: Surfaces in Three-Space

1 Name and sketch the graph of each of the following equations in three-space.

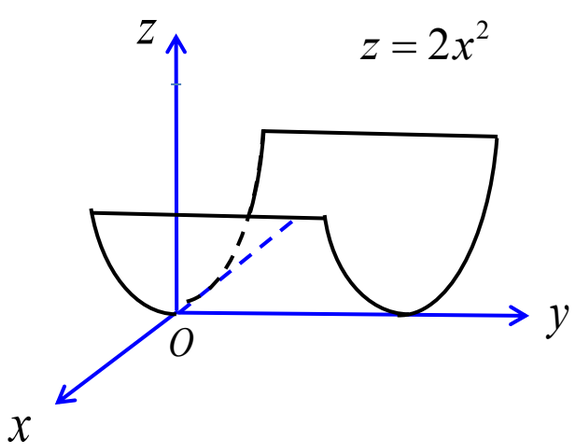
(1). $x^2+y^2=15$

Solution: The surface is a circular cylinder.



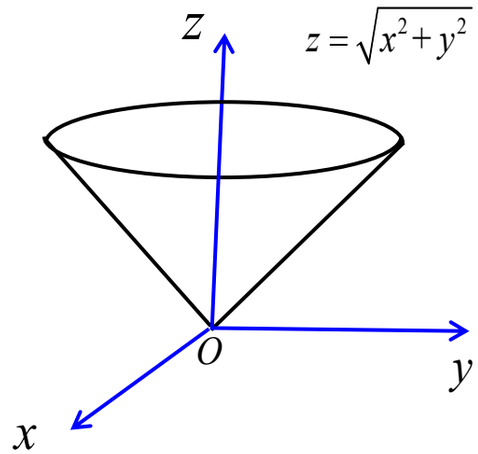
(2). $z=2x^2$

Solution: The surface is a parabolic cylinder.



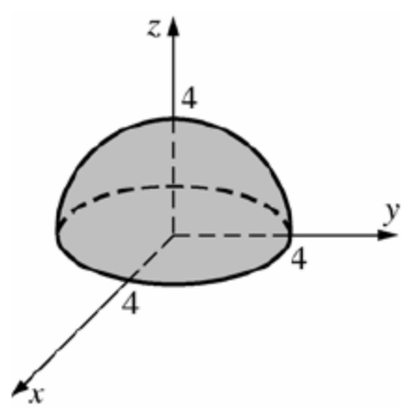
(3). $z=\sqrt{x^2+y^2}$

Solution: The surface is a circular cone.

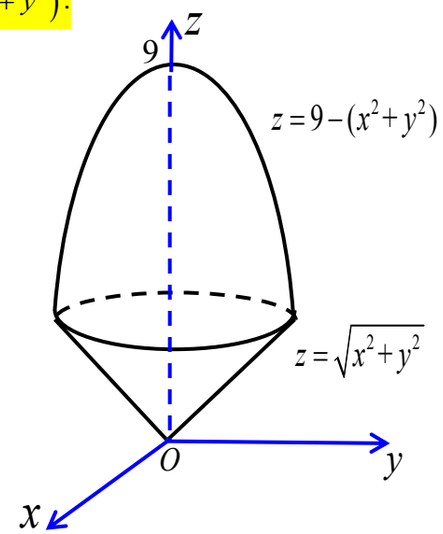


(4). $z=\sqrt{16-x^2-y^2}$

Solution: The surface is a hemisphere.



2. Sketch the region bounded by $z=\sqrt{x^2+y^2}$ and $z=9-(x^2+y^2)$.



3. If the curve $z=x^2$ in the xz-plane is revolved about the z-axis, the resulting surface has equation $z=x^2+y^2$, obtained as a result of replacing x by $\sqrt{x^2+y^2}$. If $y=2x^2$ in the xy-plane is revolved about the y-axis, what is the equation of the resulting surface?

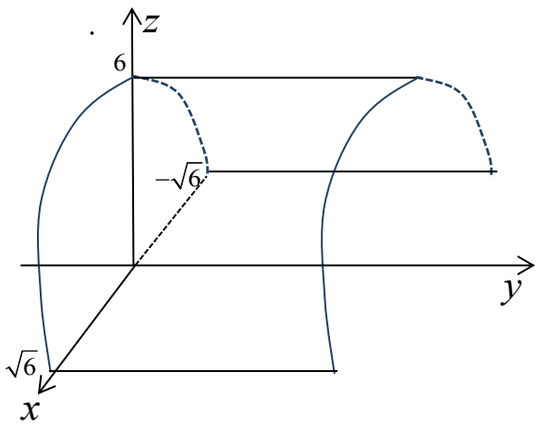
Solution: The equation of the resulting surface is $y=2x^2+2z^2$.

Exercises 12.1: Functions of Two or More Variables

1. Sketch the graph of *f*

(1). $f(x,y)=6-x^2$

Solution: The trace in the $zx-$plane is $z=6-x^2$.



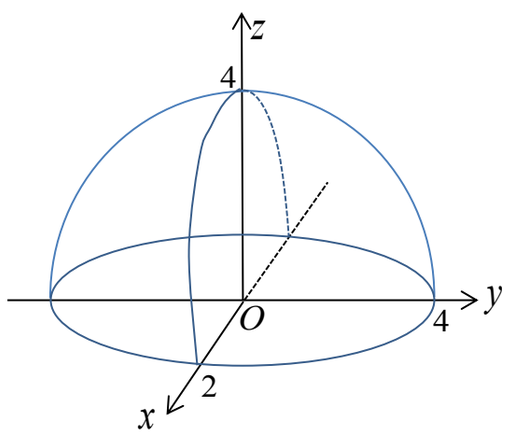
(2). $f(x,y)=\sqrt{16-4x^2-y^2}$

Solution:

The trace in the $xy-$plane has equation $4x^2+y^2=16$.

The trace in the $zx-$ plane has equation $4x^2+z^2=16\ (z\ge 0)$.

The trace in the $yz-$ plane has equation $z^2+y^2=16\ (z\ge 0)$.



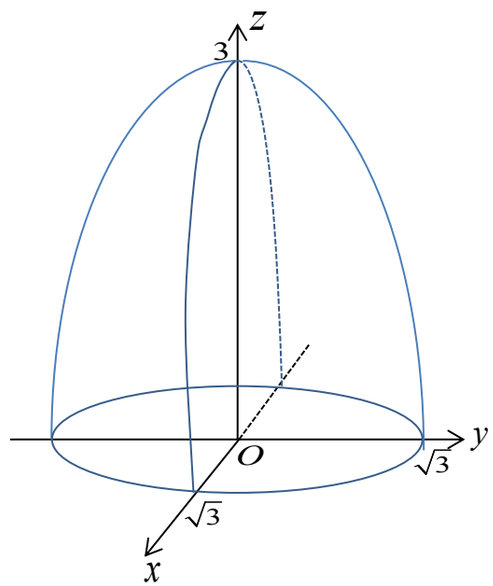
(3). $f(x,y)=3-x^2-y^2$

Solution:

The trace in the $xy-$plane has equation $x^2+y^2=3$.

The trace in the $zx-$plane has equation $z=3-x^2$.

The trace in the z plane has equation $z=3-y^2$.



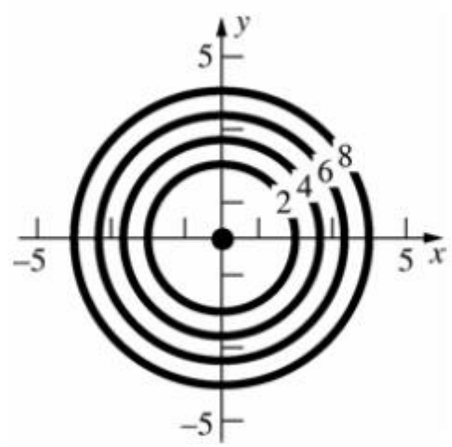
2. Sketch the level curve $z=k$ for the indicated values of *k*.

(1). $z=\frac{1}{2}\left(x^2+y^2\right),\ k=0,2,4,6,8$.

Solution:

$x^2+y^2=2z$.

Whenever $z=k=0,2,4,6,8,\quad x^2+y^2=0,4,8,12,16$.

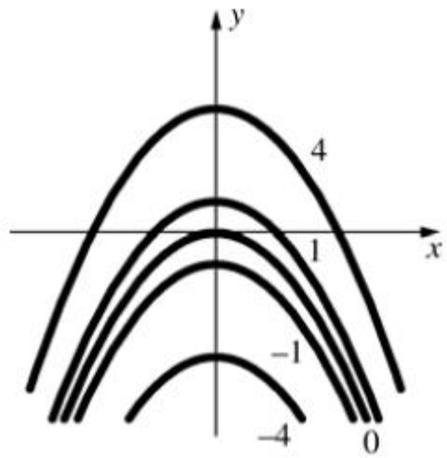


(2). $z=x^2+y, k=-4,-1,0,1,4$.

Solution:

$x^2+y=z$.

Whenever $z=k=-4,-1,0,1,4,\quad x^2+y=-4,-1,0,1,4$



3. Identify the graph of $f(x,y)=x^2-x+3y^2+12y-13$, state where it attains its minimum value, and find this minimum value.

Solution:

Completing the squares gives

$$f(x,y)=\left(x-\frac{1}{2}\right)^2+3(y+2)^2-\frac{1}{4}-12-13$$

$$=\left(x-\frac{1}{2}\right)^2+3(y+2)^2-\frac{10}{4}$$, which is an elliptic paraboloid.

Therefore, $f(x,y)$ attains its minimum value at the point $\left(\frac{1}{2},-2\right)$ and the minimum value is $f(\frac{1}{2},-2)=-\frac{101}{4}$.

Exercises 12.2: Partial Derivatives

1. Find all first partial derivatives of each function.

(1). $f(x,y)=(3x+2y^3)^{4/3}$.

Solution:

$$\frac{\partial f}{\partial x}=\frac{4}{3}(3x+2y^3)^{\frac{1}{3}}\cdot 3=4(3x+2y^3)^{\frac{1}{3}},$$

$$\frac{\partial f}{\partial y}=\frac{4}{3}(3x+2y^3)^{\frac{1}{3}}\cdot 6y^2=8y^2(3x+2y^3)^{\frac{1}{3}}.$$

(2). $f(x,y)=\dfrac{x^2-y^2}{xy}$

Solution:

$$f(x,y)=\frac{x}{y}-\frac{y}{x}.$$

$$\frac{\partial f}{\partial x}=\frac{1}{y}-y\left(-\frac{1}{x^2}\right)=\frac{x^2+y^2}{x^2y}$$

$$\frac{\partial f}{\partial y}=-\frac{x}{y^2}-\frac{1}{x}=-\frac{x^2+y^2}{xy^2}$$

(3). $f(x,y)=e^x\cos y$

Solution:

$$\frac{\partial f}{\partial x}=e^x\cos y$$

$$\frac{\partial f}{\partial y}=-e^x\sin y$$

(4). $f(x,y)=e^{-y}\sin(x^2+y^2)$.

Solution:

$$\frac{\partial f}{\partial x}=e^{-y}\cos(x^2+y^2)\cdot 2x=2xe^{-y}\cos(x^2+y^2),$$

$$\frac{\partial f}{\partial y}=-e^{-y}\sin(x^2+y^2)+2ye^{-y}\cos(x^2+y^2).$$

(5). $f(x,y)=\ln(2x+y^2)$.

Solution:

$$\frac{\partial f}{\partial x}=\frac{2}{2x+y^2},$$

$$\frac{\partial f}{\partial y}=\frac{2y}{2x+y^2}.$$

2. If $f(x,y)=\arctan\left(\dfrac{y}{x^2}\right)$, find $f_y(\sqrt{5},-2)$.

Solution:

$$f_y=\frac{1}{1+\left(\frac{y}{x^2}\right)^2}\cdot\left(\frac{1}{x^2}\right)=\frac{x^2}{x^4+y^2},$$

$$f_y(\sqrt{5},-2)=\frac{5}{25+4}=\frac{5}{29}.$$

3. If $F(x,y)=3x^4y^5-2x^2y^3$, find $\partial^2F(x,y)/\partial y^2$

Solution:

$$\frac{\partial F(x,y)}{\partial y}=3x^4\cdot 5y^4-2x^2\cdot 3y^2=15x^4y^4-6x^2y^2$$

$$\frac{\partial^2F(x,y)}{\partial y^2}=\frac{\partial}{\partial y}\frac{\partial F(x,y)}{\partial y}=60x^4y^3-12x^2y$$

4. If $f(x,y,z)=3x^2y-xyz+y^2z^2$, find each of the following:

(a) $f_x(x,y,z)$ (b) $f_y(0,1,2)$ (c) $f_{xy}(x,y,z)$

Solution:

(a) $f_x(x,y,z)=6xy-yz$

(b) $f_y(x,y,z)=3x^2-xz+2yz^2$; $f_y(0,1,2)=8$

(c) $f_{xy}(x,y,z)=6x-z$

5. The Cobb-Douglas production function for a new product is given by $P(x,y)=10x^{0.4}y^{0.6}$, where x is the number of the units of labour and y is the number of the units of capital required to produce $P(x,y)$ units of the product.

(a) Find $P_x(x,y)$ and $P_y(x,y)$.

(b) If the company is currently using 200 units of labor and 100 units of capital, find the marginal productivity of labour and the marginal productivity of capital respectively.

(c) For the greatest increase in productivity, should the management of the company encourage increased use of labour or increased use of capital?

Solution: (a) $P_x(x,y)=10\cdot 0.4x^{-0.6}y^{0.6}=4x^{-0.6}y^{0.6}$

$P_y(x,y)=10\cdot 0.6x^{0.4}y^{-0.4}=6x^{0.4}y^{-0.4}$.

(b) If the company is using 100 units of labour and 200 units of capital, then the marginal productivity of labour is

$P_x(200,100)=4\left(\dfrac{100}{200}\right)^{0.6}=4\cdot 0.5^{0.6}\approx 2.639$ and the

marginal productivity of capital is

$P_y(200,100)=6\left(\dfrac{200}{100}\right)^{0.4}=6\cdot 2^{0.4}\approx 7.917.$

(c) The management of the company should encourage increased use of capital, because $P_x(200,100)<P_y(200,100)$.

Exercises 12.3: Limits and Continuity

1. Find the indicated limit or state that it does not exist.

(1) $\lim\limits_{(x,y)\to(2,-1)}\left(x^2y+3x\right)$

Solution:

$\lim\limits_{(x,y)\to(2,-1)}\left(x^2y+3x\right)=4\cdot(-1)+6=2$.

(2) $\lim\limits_{(x,y)\to(0,0)}\dfrac{\sin(x+y)}{3(x+y)}$

Solution:

As $(x,y)\to(0,0)$, $\lim\limits_{(x,y)\to(0,0)}(x+y)=0$.

Let $u=x+y$, then

$\lim\limits_{(x,y)\to(0,0)}\dfrac{\sin(x+y)}{3(x+y)}=\lim\limits_{u\to0}\dfrac{\sin u}{3u}=\dfrac{1}{3}$.

(3). $\lim\limits_{(x,y)\to(0,0)}xy\dfrac{x^2-y^2}{x^2+y^2}$

Solution:

$$\left|xy\frac{x^2-y^2}{x^2+y^2}\right|=\left|x^2-y^2\right|\left|\frac{xy}{x^2+y^2}\right|\le\frac{1}{2}\left|x^2-y^2\right|\le\frac{1}{2}\left(x^2+y^2\right).$$

If (x,y) goes to $(0,0)$, then x^2+y^2 goes to zero, so

$\lim\limits_{(x,y)\to(0,0)}xy\dfrac{x^2-y^2}{x^2+y^2}=0$.

.

2. Show that $\lim\limits_{(x,y)\to(0,0)}\dfrac{xy+y^3}{x^2+y^2}$ does not exist.

Solution:

Along $y=0$, $\lim\limits_{x\to0}\dfrac{0}{x^2+0}=0$

Along $y=x$, $\lim\limits_{x\to0}\dfrac{x^2+x^3}{x^2+x^2}=\lim\limits_{x\to0}\dfrac{1+x}{2}=\dfrac{1}{2}$

Hence, the limit does not exists.

3. Let $f(x,y)=x^2y/\left(x^4+y^2\right)$.

(a) Show that $f(x,y)$ goes to zero if (x,y) goes to $(0,0)$ along any straight line $y=mx$.

Solution:

$$\lim_{\substack{(x,y)\to(0,0)\\ y=mx}}f(x,y)=\lim_{x\to0}\frac{x^2\cdot mx}{\left(x^4+m^2x^2\right)}=\lim_{x\to0}\frac{mx}{\left(x^2+m^2\right)}=0.$$

(b) Show that $f(x,y)$ goes to $\dfrac{1}{2}$ if (x,y) goes to $(0,0)$ along the parabola $y=x^2$.

Solution:

$$\lim_{\substack{(x,y)\to(0,0)\\ y=x^2}}f(x,y)=\lim_{x\to0}\frac{x^2\cdot x^2}{\left(x^4+x^4\right)}=\frac{1}{2}.$$

(c) What conclusion do you draw?

Solution:

The limit $\lim\limits_{(x,y)\to(0,0)}\dfrac{x^2y}{x^4+y^2}$ does not exist, because we obtain different values along different paths even if $f(x,y)$ approaches zero as (x,y) goes to $(0,0)$ along all straight lines.

Exercises 12.4: Differentiability

1. Find the gradient ∇f.

(1). $f(x,y,z)=x^2+y^3+z^5$.

Solution:

$f_x=2x$, $f_y=3y^2$, $f_z=5z^4$.

$\nabla f=\langle 2x,3y^2,5z^4\rangle$.

(2). $f(x,y,z)=xz\ln(x+y+z)$

Solution:

$f_x=z\ln(x+y+z)+\dfrac{xz}{x+y+z}$;

$f_y=\dfrac{xz}{x+y+z}$; $f_z=x\ln(x+y+z)+\dfrac{xz}{x+y+z}$;

$$\nabla f=\left\langle z\ln(x+y+z)+\frac{xz}{x+y+z},\frac{xz}{x+y+z},x\ln(x+y+z)+\frac{xz}{x+y+z}\right\rangle$$

(3) $f(x,y,z)=\sin(\mathrm{xyz})+\cos(\mathrm{z})$.

Solution:

$f_x=yz\cos(xyz)$, $f_y=xz\cos(xyz)$,

$f_z=xy\cos(xyz)-\sin(\mathrm{z})$.

$\nabla f=\langle yz\cos(xyz),xz\cos(xyz),xy\cos(xyz)-\sin(\mathrm{z})\rangle$.

2. Find the gradient vector of the given function at the given point **p**. Then find the equation of the tangent plane at **p**.

(1). $f(x,y)=x^2+y^2+4xy$ and $\mathbf{p}=(3,-1)$.

Solution:

$f(3,-1)=9+1-12=-2$,

$\nabla f=\langle 2x+4y,2y+4x\rangle$ and $\nabla f(3,-1)=\langle 2,10\rangle$.

The tangent plane at **p** is

$z=f(3,-1)+\nabla f(3,-1)\cdot\langle x-3,y+1\rangle$

$=-2+2(x-3)+10(y+1)$

that is, $z=2x+10y+2$.

(2). $f(x,y)=\dfrac{x^2}{y}$, $\mathbf{p}=(2,-1)$

Solution:

$f(2,-1)=\dfrac{2^2}{-1}=-4$; $\nabla f=\left\langle\dfrac{2x}{y},-\dfrac{x^2}{y^2}\right\rangle$,

$$\nabla f(2,-1)=\left\langle\frac{4}{-1},-\frac{2^2}{(-1)^2}\right\rangle=\langle -4,-4\rangle.$$

The tangent plane at **p** is

$z=f(2,-1)+\nabla f(2,-1)\cdot\langle x-2,y+1\rangle$

$=-4-4(x-2)-4(y+1)$

i.e. $z=-4x-4y$.

3. Find all points (x,y) for which the tangent plane to the graph of $z=x^2-6x+2y^2-10y+2xy$ is horizontal.

Solution:

The normal vector to the plane tangent to the graph at $(x,y,f(x,y))$ is

$\vec{n}=\langle f_x,f_y,-1\rangle=\langle 2x-6+2y,4y-10+2x,-1\rangle$.

The tangent plane is horizontal if and only if the normal vector can be written as $\langle 0,0,k\rangle$,where $k\neq 0$, i.e.

$2x-6+2y=0$, $4y-10+2x=0$ and $k=-1$. Solving for x and y we get $x=1$ and $y=2$. .

Therefore, there is a horizontal tangent plane at the point $(1,2)$.

4. Find parametric equations of the line tangent to the surface $z=y^2+x^3y$ at the point (2,1,9) whose projection on the xy-plane is parallel to the x-axix.

Solution1 By the given condition, the tangent line can be regarded as the intersection line of the plane y =1 and the tangent plane to the surface $z=y^2+x^3y$ at the point (2,1,9).

The equation of the tangent plane is

$z=f(2,1)+\nabla f(2,1)\cdot\langle x-2,y-1\rangle$

$=9+\langle 12,10\rangle\cdot\langle x-2,y-1\rangle=12x+10y-25$

So the tangent line has the direction vector

$\mathbf{v}=\langle 12,10,-1\rangle\times\langle 0,1,0\rangle=\langle 1,0,12\rangle$.

Thus, the parametric equations for the tangent line are

$$\begin{cases}x=2+t\\ y=1\\ z=9+12t\end{cases}$$

Solution 2 The point (2,1,9) projects to (2,1,0) on the xy-plane. The equation of a plane containing this point and parallel to the x-axis is given by y = 1.

Let C be the intersection curve of the surface $z=y^2+x^3y$ and the plane y = 1. Then the tangent line touches C at the point (2,1,9) on the plane y = 1.

The vector equation of C is $\mathbf{r}(x)=\langle x,1,1+x^3\rangle$,

$\mathbf{r}'(x)=\langle 1,0,3x^2\rangle$.

The direction vector of the tangent line is $\mathbf{r}'(2)=\langle 1,0,12\rangle$.

Thus, parametric equations for the desired tangent line are

$$\begin{cases}x=2+t\\ y=1\\ z=9+12t\end{cases}$$

Exercises 12.5: Directional Derivatives and Gradients

1. Find the directional derivative of f at the point **p** in the direction of **a**.

(1). $f(x,y)=xy, \mathbf{p}=(1,1)$ and $\vec{a}=\left\langle \frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}} \right\rangle$.

Solution:

$\nabla f(x,y)=\langle y,x\rangle$ and $\vec{a}$ is a unit vector. So

$$D_{\vec{a}}f(x,y)=\nabla f(x,y)\cdot\vec{a}$$
$$=\langle y,x\rangle\cdot\left\langle \frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}} \right\rangle$$
$$=\frac{1}{\sqrt{2}}(x+y).$$

Therefore, $D_{\vec{u}}f(1,1)=\sqrt{2}$.

(2). $f(x,y,z)=x^3y-y^2z^2;\ \mathbf{p}=(-2,1,3);\ \mathbf{a}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$

Solution:

$\nabla f(x,y,z)=\langle 3x^2y, x^3-2yz^2, -2y^2z\rangle$

The unit vector in the direction $\mathbf{a}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$ is

$$\mathbf{u}=\frac{1}{\sqrt{1^2+(-2)^2+2^2}}\langle 1,-2,2\rangle=\left\langle \frac{1}{3},-\frac{2}{3},\frac{2}{3} \right\rangle, \text{ so}$$

$$D_{\mathbf{u}}f(x,y,z)=\nabla f(x,y,z)\cdot\mathbf{u}$$
$$=\langle 3x^2y, x^3-2yz^2, -2y^2z\rangle\cdot\left\langle \frac{1}{3},-\frac{2}{3},\frac{2}{3} \right\rangle$$
$$=x^2y-\frac{2}{3}(x^3-2yz^2)-\frac{4}{3}y^2z,$$

giving $D_{\mathbf{u}}f(-2,1,3)=\frac{52}{3}$.

2. Find a unit vector in the direction in which *f* increases most rapidly at **p**. What is the rate of change in this direction?

$f(x,y,z)=xe^{yz};\ \mathbf{p}=(2,0,-4)$

Solution:

f increases most rapidly in the direction of the gradient.

$\nabla f(x,y,z)=\langle e^{yz}, xze^{yz}, xye^{yz}\rangle$, so

$\nabla f(2,0,-4)=\langle 1,-8,0\rangle$.

The unit vector with direction $\langle 1,-8,0\rangle$ is

$$\frac{1}{\sqrt{1^2+(-8)^2+0^2}}\langle 1,-8,0\rangle=\frac{\langle 1,-8,0\rangle}{\sqrt{65}}.$$

The rate of change in this direction is $\|\langle 1,-8,0\rangle\|=\sqrt{65}$.

3. Find the directional derivative of $f(x,y,z)=xy+z^2$ at $(1,1,1)$ in the direction towards $(5,-3,3)$.

Solution:

The vector from (1,1,1) to (5,-3,3) is
$\mathbf{a}=\langle 5-1,-3-1,3-1\rangle=\langle 4,-4,2\rangle=2\langle 2,-2,1\rangle$.

The unit vector in the direction $\mathbf{a}$ is

$$\mathbf{u}=\frac{1}{\sqrt{2^2+(-2)^2+1^2}}\langle 2,-2,1\rangle=\left\langle \frac{2}{3},-\frac{2}{3},\frac{1}{3} \right\rangle.$$

As $\nabla f(x,y,z)=\langle y,x,2z\rangle$ and $\nabla f(1,1,1)=\langle 1,1,2\rangle$, we get that the directional derivative of $f(x,y,z)$ at (1,1,1) in the direction towards (5,-3,3) is $D_{\mathbf{u}}f(1,1,1)=\frac{2}{3}$.

4. The temperature at (x,y,z) of a solid sphere centered at the origin is given by $T(x,y,z)=\frac{200}{5+x^2+y^2+z^2}$

(a) By inspection, decide where the solid sphere is hottest.
(b) Find the unit vector pointing in the direction of greatest increase of temperature at $(1,-1,1)$.
(c) Find the unit vector pointing in the direction of greatest decrease of temperature at $(1,-1,1)$.

Solution:

(a). Hottest if denominator is smallest; i.e., at the origin.

(b)

$$\nabla T(x,y,z)=\frac{-200\langle 2x,2y,2z\rangle}{(5+x^2+y^2+z^2)^2};$$

$$\nabla T(1,-1,1)=-\frac{25}{4}\langle 1,-1,1\rangle$$

Hence, $\frac{\sqrt{3}}{3}\langle -1,1,-1\rangle$ is the unit vector in the direction of greatest increase at $(1,-1,1)$.

(c). From (b), $-\frac{\sqrt{3}}{3}\langle -1,1,-1\rangle$ is the unit vector in the direction of greatest decrease at $(1,-1,1)$.

Exercises 12.6: The Chain Rule

1. Find dw/dt by using the Chain Rule. Express your final answer in terms of *t*.

$$w = x^2y - y^2x;\ x = \cos t, y = \sin t$$

Solution:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\partial w}{\partial x}\cdot\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial w}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}t}$$

$$= \left(2xy - y^2\right)\left(-\sin t\right) + \left(x^2 - 2xy\right)\cos t$$

$$= \left(2\cos t\sin t - \sin^2 t\right)\left(-\sin t\right) + \left(\cos^2 t - 2\cos t\sin t\right)\cos t$$

$$= \sin^3 t + \cos^3 t + \left(-2\cos t\sin^2 t - 2\cos^2 t\sin t\right)$$

$$= \left(\sin t + \cos t\right)\left(1 - 3\sin t\cos t\right).$$

2. Find $\partial w/\partial t$ by using the Chain Rule. Express your final answer in terms of *s* and *t*.

$$w = e^{xy+z};\ x = s+t, y = s-t, z = t^2$$

Solution:

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\cdot\frac{\mathrm{d}z}{\mathrm{d}t}$$

$$= \mathrm{e}^{xy+z}\, y\cdot 1 + \mathrm{e}^{xy+z}\, x\cdot(-1) + \mathrm{e}^{xy+z}\cdot 2t$$

$$= \mathrm{e}^{xy+z}\left(y - x + 2t\right)$$

$$= \mathrm{e}^{s^2}\left(-2t + 2t\right) = 0.$$

3. If $z = x^2y, x = 2t+s$, and $y = 1 - st^2$, find $\left.\frac{\partial z}{\partial t}\right|_{s=1,t=2}$.

Solution:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial t}$$

$$= 2xy\cdot 2 + x^2\cdot(-2st)$$

$$= 4\left(2t+s\right)\left(1 - st^2\right) + \left(2t+s\right)^2\left(-2st\right),\ \text{so}$$

$$\left.\frac{\partial z}{\partial t}\right|_{s=1,\,t=2} = 4\left[2\cdot 2+1\right]\left[1 - 1\cdot 2^2\right]$$

$$+\left[2\cdot 2+1\right]^2\left[-2\cdot 1\cdot 2\right] = -160.$$

4. Find *dy*/*dx*, where y(x) is an implicit function defined by $x^2\cos y - y^2\sin x = 0$

Solution:

Let $F\left(x, y\right) = x^2\cos y - y^2\sin x = 0$. Then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y} = -\frac{\left(2x\cos y - y^2\cos x\right)}{-x^2\sin y - 2y\sin x} = \frac{2x\cos y - y^2\cos x}{x^2\sin y + 2y\sin x}.$$

5. If $3x^2z + y^3 - xyz^3 = 0$, find $\partial z/\partial x$.

Solution:

Let $F\left(x, y, z\right) = 3x^2z + y^3 - xyz^3 = 0$. Then

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{6xz - yz^3}{3x^2 - 3xyz^2} = \frac{yz^3 - 6xz}{3x^2 - 3xyz^2}.$$

6. Show that if $w = f(r-s, s-t, t-r)$, then $\frac{\partial w}{\partial r} + \frac{\partial w}{\partial s} + \frac{\partial w}{\partial t} = 0$.

Solution:

Let $x = r-s,\ y = s-t,\ z = t-r$. Then

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial r}$$

$$= f_x\cdot 1 + f_z\cdot(-1) = f_x - f_z,$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial s}$$

$$= f_x\cdot(-1) + f_y\cdot 1 = -f_x + f_y \text{ and}$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial t}$$

$$= f_y\cdot(-1) + f_z\cdot 1 = -f_y + f_z.$$

Therefore

$$\frac{\partial w}{\partial r} + \frac{\partial w}{\partial s} + \frac{\partial w}{\partial t} = f_x - f_z - f_x + f_y - f_y + f_z = 0.$$

Exercise 12.7: Tangent Planes and Approximations

1. Find the equation of the tangent plane to the given surface at the indicated point.

(1). $x^2+y^2-z^2=4;\ (2,1,1)$

Solution:

Let $F(x,y,z)=x^2+y^2-z^2-4$. Then

$$F_x=2x,\ F_y=2y \text{ and } F_z=-2$$

so the tangent plane at (2,1,1) has normal vector $\vec{n}=\nabla F(2,1,1)=\langle F_x,F_y,F_z\rangle\big|_{(2,1,1)}=\langle 4,2,-2\rangle=2\langle 2,1,-1\rangle$

and its equation is

$$2(x-2)+1(y-1)-1(z-1)=0$$

or $$2x+y-z=4.$$

(2). $z=x^{1/2}+y^{1/2};\ (1,4,3)$

Solution:

Let $F(x,y,z)=x^{\frac{1}{2}}+y^{\frac{1}{2}}-z.$

Then $F_x=\frac{1}{2}x^{-\frac{1}{2}},\ F_y=\frac{1}{2}y^{-\frac{1}{2}}$ and $F_z=-1,$

so the tangent plane at (1,4,3) has normal vector $\vec{n}=\nabla F(1,4,3)=\langle F_x,F_y,F_z\rangle\big|_{(1,4,3)}=\left\langle\frac{1}{2},\frac{1}{4},-1\right\rangle$

and the tangent plane at $(1,4,3)$ has equation $\frac{1}{2}(x-1)+\frac{1}{4}(y-4)+(-1)(z-3)=0,$ or $2x+y-4z=-6$.

2. Find a point on the surface $z=2x^2+3y^2$ where the tangent plane is parallel to the plane $8x-3y-z=0$.

Solution:

Let the coordinates of the point be (x_0,y_0,z_0) and let $F(x,y,z)=2x^2+3y^2-z$.

Then the tangent plane at point (x_0,y_0,z_0) has normal vector $\vec{n}=\nabla F(x_0,y_0,z_0)=\langle 4x_0,6y_0,-1\rangle$.

Since the tangent plane is parallel to plane $8x-3y-z=0$, so the normal vectors of the two planes are parallel to each other, that is $\langle 4x_0,6y_0,-1\rangle//\langle 8,-3,-1\rangle$. Thus $\frac{4x_0}{8}=\frac{6y_0}{-3}=\frac{-1}{-1}$, meaning that $x_0=2$ and $y_0=-\frac{1}{2}$.

Additionally, $z_0=2x_0^2+3y_0^2=8\frac{3}{4}$. Therefore, at point $\left(2,-\frac{1}{2},8\frac{3}{4}\right)$, the tangent plan to the surface $z=2x^2+3y^2$ is parallel to $8x-3y-z=0$.

3. Show that the surface S_1: $x^2+4y+z^2=0$ and the surface S_2: $x^2+y^2+z^2-6z+7=0$ are tangent to each other at the point P(0,-1,2); that is, show that they have the same tangent plane at (0,-1,2).

Proof: Let $F(x,y,z)=x^2+4y+z^2$ and

$$G(x,y,z)=x^2+y^2+z^2-6z+7$$

Since $F(0,-1,2)=0^2+4(-1)+2^2=0$ and $G(0,-1,2)=0^2+(-1)^2+2^2-6\times 2+7=0$, we know that point P lies on both surfaces.

The tangent plane to surface S_1 at point P has normal vector

$\vec{n_1}=\nabla F(0,-1,2)=\langle F_x,F_y,F_z\rangle\big|_{(0,-1,2)}=\langle 2x,4,2z\rangle\big|_{(0,-1,2)}=\langle 0,4,4\rangle.$

The tangent plane to surface S_2 at point P has normal vector $\vec{n_2}=\nabla G(0,-1,2)=\langle G_x,G_y,G_z\rangle\big|_{(0,-1,2)}$

$$=\langle 2x,2y,2z-6\rangle\big|_{(0,-1,2)}=\langle 0,-2,-2\rangle,$$

Since $\vec{n_1}//\vec{n_2}$, two surface have the same tangent plane at point P.

4. Find all points on the surface S: $x^2+2y^2+3z^2=12$ where the tangent plane is perpendicular to the line L: $x=1+2t, y=3+8t, z=2-6t.$

Solution: Let $F(x,y,z)=x^2+2y^2+3z^2-12$. Then the tangent plane to the surface at point (x,y,z) has normal vector $\vec{n}=\nabla F(x,y,z)=\langle 2x,4y,6z\rangle=2\langle x,2y,3z\rangle$.

Since the tangent plane is perpendicular to the line L of which a direction vector is $\langle 2,8,-6\rangle$, we have $\vec{n}//\langle 2,8,-6\rangle$,

meaning $2\langle x,2y,3z\rangle=\lambda\langle 2,8,-6\rangle$, where λ is a scalar to be determined. We thus get that $x=\lambda$, $y=2\lambda$ and $z=-\lambda$.

The point (x,y,z) lies on the surface $x^2+2y^2+3z^2=12$, so $\lambda^2+2\cdot(2\lambda)^2+3\cdot(-\lambda)^2=12$, giving $\lambda=\pm 1$. We therefore conclude that the points on the surface where the tangent plane is perpendicular to the line with parametric equations $x=1+2t$, $y=3+8t$ and $z=2-6t$ are $(1,2,-1)$ (for λ=1) and $(-1,-2,1)$ (as $\lambda=-1$).

5. Use the total differential dz to approximate the change in z as (x,y) moves from P to Q. Then use a calculator to find the corresponding exact change Δz.

$$z=x^2-5xy+y;\ P(2,3),\ Q(2.03,2.98)$$

Solution:

$dz=(2x-5y)dx+(-5x+1)dy$

Substituting $(x,y)=(2,3),(\Delta x,\Delta y)=(0.03,-0.02)$ into the above formula, we have

$dz=(-11)(0.03)+(-9)(-0.02)=-0.15$

$\Delta z=f(2.03,2.98)-f(2,3)=-1.1461$

Exercises 12.8: Maxima and Minima

1. Find all critical points. Indicate whether each such point gives a local maximum or a local minimum, or whether it is a saddle point.

(1). $f(x,y)=2x^4-x^2+3y^2$

Solution:

$f(x,y)$ is differentiable in the whole domain, so any critical points have to be stationary points. Solving $f_x=8x^3-2x=0$ and $f_y=6y=0$ gives $y=0$ and $x=0,\pm\frac{1}{2}$, so $(0,0),\left(\frac{1}{2},0\right)$ and $\left(-\frac{1}{2},0\right)$ are the stationary points.

$f_{xx}=24x^2-2,\ f_{yy}=6,\ f_{xy}=0$ and

$D=f_{xx}f_{yy}-f_{xy}^2=12\left(12x^2-1\right)$.

At $(0,0)$, $D=-12$, so $(0,0)$ is a saddle point.

At $\left(\frac{1}{2},0\right)$, $D=24$ and $f_{xx}=4$, so $f\left(\frac{1}{2},0\right)$ is a local minimum.

At $\left(-\frac{1}{2},0\right)$, $D=24$ and $f_{xx}=4$, so $f\left(-\frac{1}{2},0\right)$ is also a local minimum.

(2). $f(x,y)=xy+\frac{2}{x}+\frac{4}{y}$

Solution:

Solving $f_x=y+2\left(-x^{-2}\right)=0$ and $f_y=x+4\left(-y^{-2}\right)=0$ gives $x=1$ and $y=2$.

$f_{xx}=-2\left(-2x^{-3}\right)=4x^{-3}$.

$f_{yy}=-4\left(-2y^{-3}\right)=8y^{-3}$.

$f_{xy}=1$.

$D=f_{xx}\cdot f_{yy}-f_{xy}^2=32x^{-3}y^{-3}-1$.

At $(1,2)$, $D=3>0,\ f_{xx}>0$, so $f(1,2)$ is a local minimum.

2. Find the global maximum and global minimum of f on S and indicate where each occurs.

(1). $f(x,y)=x^2+y^2;\ S=\{(x,y):-1\le x\le 3,-1\le y\le 4\}$

Solution: Solving $f_x=2x=0$ and $f_y=2y=0$ shows that $(0,0)$ is the only stationary point of f inside S.

$$f(0,0)=0^2+0^2=0$$

On the boundary of S, f reaches minimum at (0,-1) and (-1, 0) with value $f(0,-1)=f(-1,0)=1$, and reaches maximum at (3, 4) with value $f(3,4)=3^2+4^2=25$.

To summarize, the global maximum is 25 and occurs at $(3,4)$ and the global minimum is 0 and occurs at $(0,0)$.

(2). $f(x,y)=x^2-6x+y^2-8y+7;\ S=\{(x,y):x^2+y^2\le 1\}$

Solution: Solving $f_x=2x-6=0$ and $f_y=2y-8=0$ gives $x=3$ and $y=4$. However, $(3,4)\notin S$, so there is no stationary point inside S and no singular point either. On the boundary of S, let $x=\cos t,\ y=\sin t,\ t\in[0,2\pi]$. Then

$$f(x,y)=f(x(t),y(t))=\cos^2 t-6\cos t+\sin^2 t-8\sin t+7$$
$$=8-6\cos t-8\sin t=:F(t).$$

To find the extreme values of $F(t)$, solve $F'(t)=6\sin t-8\cos t=0$, which gives $\tan t=\frac{4}{3}$. This means that the corresponding t lies in the interval $\left(0,\frac{\pi}{2}\right)$, or the interval $\left(\pi,\frac{3\pi}{2}\right)$. The corresponding points on S are $\left(\frac{3}{5},\frac{4}{5}\right)$ and $\left(-\frac{3}{5},-\frac{4}{5}\right)$. As F is defined on the closed interval $[0,2\pi]$, we also need to check whether F takes on a maximum, or minimum at the boundary points $t=0$ and $t=2\pi$ of that interval: We find that $F(0)=F(2\pi)=8-6-0=2$.

As $f\left(\frac{3}{5},\frac{4}{5}\right)=-2$, $f\left(-\frac{3}{5},-\frac{4}{5}\right)=18$ and $f(1,0)=F(0)=F(2\pi)=2$, we find that the global maximum of f is 18 and reached at $\left(-\frac{3}{5},-\frac{4}{5}\right)$ and that the global minimum of f is -2 and reached at $\left(\frac{3}{5},\frac{4}{5}\right)$.

3. Find the three-dimensional vector with length 9, the sum of whose components is a maximum.

Solution:

Let $\langle x,y,z\rangle$ denote the vector; let S be the sum of its components.

$x^2+y^2+z^2=81$, so $z=\left(81-x^2-y^2\right)^{1/2}$.

Maximize $S(x,y)=x+y+\left(81-x^2-y^2\right)^{1/2},\ x^2+y^2\le 81$.

Let

$$\nabla S(x,y)=\left\langle 1-x\left(81-x^2-y^2\right)^{-1/2},1-y\left(81-x^2-y^2\right)^{-1/2}\right\rangle=\langle 0,0\rangle$$

Therefore, $x=\left(81-x^2-y^2\right)^{1/2}$ and $y=\left(81-x^2-y^2\right)^{1/2}$.

We then obtain $x=y=3\sqrt{3}$ as the only stationary point. For these values of x and y, $z=3\sqrt{3}$ and $S=9\sqrt{3}\approx 15.59$.

On the boundary $x^2+y^2=81$, let $x=9\cos t,y=9\sin t,0\le t\le 2\pi$. Let $f(t)=S(9\cos t,9\sin t)=9\cos t+9\sin t,0\le t\le 2\pi$.

$f'(t)=0\Rightarrow t_1=\pi/4,t_2=5\pi/4$. Adding two end points $t_3=0,t_4=2\pi$. Comparing $f(t_k),k=1,2,3,4$, We get the max value on the boundary is $f(t_1)=9\sqrt{2}<9\sqrt{3}$..

Thus we conclude that the required vector is $3\sqrt{3}\langle 1,1,1\rangle$.

Exercises 12.9: The Method of Lagrange Multipliers

1. Find the maximum of $f(x,y)=4x^2-4xy+y^2$ subject to the constraint $x^2+y^2=1$.

Solution: Let $g(x,y)=x^2+y^2-1$.

$\nabla f(x,y)=\langle 8x-4y,-4x+2y\rangle$, $\nabla g(x,y)=\langle 2x,2y\rangle$.

The Lagrange equations are:

$$8x-4y=2\lambda x \quad (1)$$
$$-4x+2y=2\lambda y \quad (2)$$
$$x^2+y^2=1. \quad (3)$$

Multiplying both sides of (1) by y and both sides of (2) by x gives $4xy-2y^2=-2x^2+xy=\lambda xy$

$\Leftrightarrow 2x^2+3xy-2y^2=0 \Leftrightarrow y=2x$ or $y=-\frac{x}{2}$.

Plugging into (3), we obtain four critical points

$\left(\frac{\sqrt5}{5},\frac{2\sqrt5}{5}\right)$, $\left(-\frac{\sqrt5}{5},-\frac{2\sqrt5}{5}\right)$, $\left(\frac{2\sqrt5}{5},-\frac{\sqrt5}{5}\right)$, $\left(-\frac{2\sqrt5}{5},\frac{\sqrt5}{5}\right)$.

$f\left(\frac{\sqrt5}{5},\frac{2\sqrt5}{5}\right)=f\left(-\frac{\sqrt5}{5},-\frac{2\sqrt5}{5}\right)=0$, and

$f\left(\frac{2\sqrt5}{5},-\frac{\sqrt5}{5}\right)=f\left(-\frac{2\sqrt5}{5},\frac{\sqrt5}{5}\right)=5$.

so the maximum of f is 5.

2. Find the maximum volume of the rectangular box in the first octant with faces parallel to the coordinate planes, one vertex at (0,0,0), and diagonally opposite vertex in the plane with equation $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$.

Solution: Let v be the volume of the box. Finding the maximum value of the rectangular box is equivalent with maximizing $v(x,y,z)=xyz$ subject to $g(x,y,z)=\frac{x}{a}+\frac{y}{b}+\frac{z}{c}-1=0$. We therefore need to solve

$\nabla v=\langle yz,xz,xy\rangle=\lambda\cdot\nabla g=\lambda\cdot\left\langle\frac{1}{a},\frac{1}{b},\frac{1}{c}\right\rangle$, or equivalent

$$yz=(1/a)\lambda \quad (1)$$
$$xz=(1/b)\lambda \quad (2)$$
$$xy=(1/c)\lambda \quad (3)$$

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1. \quad (4)$$

Multiplying both sides of (1) by x and both sides of (2) by y gives $\frac{1}{a}\lambda x=xyz=\frac{1}{b}\lambda y$. As the volume of the box is nonzero, neither x, y, nor z can be equal to zero, so because of (1), $\lambda\neq 0$, which means that $\frac{1}{a}x=\frac{1}{b}y$.

Multiplying (3) by z and using again that $xyz=\frac{1}{c}\lambda z$ gives $\frac{1}{c}z=\frac{1}{b}y$. So (4) together with $\frac{1}{a}x=\frac{1}{b}y=\frac{1}{c}z$ gives that

$$1=\frac{1}{a}x+\frac{1}{b}y+\frac{1}{c}z=\frac{1}{a}x+\frac{1}{a}x+\frac{1}{a}x$$
$$=\frac{1}{b}y+\frac{1}{b}y+\frac{1}{b}y=\frac{1}{c}z+\frac{1}{c}z+\frac{1}{c}z,$$

which means that $\frac{1}{c}z=\frac{1}{a}x=\frac{1}{b}y=\frac{1}{3}$. So the only critical point of v is $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$ and consequently the maximum of v is $v=\frac{abc}{27}$.

3. Find the global maximum value and global minimum value of f on the closed and bounded set S and indicate where each occurs. (Use Lagrange multipliers to find the maximum and minimum on the boundary of S).

$f(x,y)=10+x+y;\ S=\{(x,y):x^2+y^2\le 1\}$

Solution:

As $f_x=1>0$ and $f_y=1>0$, there are no critical points in the interior of S.

To find the global extremes on the boundary of S, let $g(x,y)=x^2+y^2-1=0$. Then solving

$\nabla f(x,y)=\langle 1,1\rangle=\lambda\nabla g(x,y)=\lambda\langle 2x,2y\rangle$, or equivalently,

solving $\begin{cases}1=\lambda\cdot 2x\\1=\lambda\cdot 2y\\x^2+y^2=1,\end{cases}$ gives $x=y=\pm\frac{1}{\sqrt2}$. So the critical points are $\left(\frac{1}{\sqrt2},\frac{1}{\sqrt2}\right)$ and $\left(-\frac{1}{\sqrt2},-\frac{1}{\sqrt2}\right)$. Therefore,

$f_{\max}=f\left(\frac{1}{\sqrt2},\frac{1}{\sqrt2}\right)=10+\sqrt2, f_{\min}=f\left(-\frac{1}{\sqrt2},-\frac{1}{\sqrt2}\right)=10-\sqrt2$

4. The Cobb-Douglas production function for a new product is given by $z=P(x,y)=9x^{\frac13}y^{\frac23}$, where x is the number of the units of labor and y is the number of the units of capital required to produce $P(x,y)$ units of the product. Each unit of labor costs 50RMB and each unit of capital costs 200 RMB.

If 300000 RMB has been budgeted for the production of this product, how should this amount be allocated in order to maximize the amount of the products? What is the maximum amount?

Solution: Maximize $P(x,y)=9x^{\frac13}y^{\frac23}$, subject to $g(x,y)=50x+200y-300,000=0$:

We need to solve $\nabla P(x,y)=\lambda\nabla g(x,y)$, or equivalently

$3x^{-\frac23}y^{\frac23}=50\lambda$ (1), $6x^{\frac13}y^{-\frac13}=200\lambda$ (2), $50x+200y=300,000$ (3)

x and y are positive and by (1) we have that λ is not zero either, so dividing the left hand side and the right hand side of (2) by the left hand side and the right hand side of (1) respectively, gives $x=2y$, which allows us to rewrite (3) as $300,000=50x+200y=100y+200y=300y$, giving $y=1000$ and $x=2000$. Since $P(x,y)$ has only one critical point, we conclude that $P(x,y)$ takes on its maximum for $x=2000$ and $y=1000$, so the maximum amount of the products is $P(2000,1000)=9(2000)^{\frac13}(1000)^{\frac23}=9000\sqrt[3]{2}$.

Exercise 13-1 Double integrals over rectangles

1. Use the property of integrals to evaluate the following integrals.

Let $R=\{(x,y):0\le x\le 2, 0\le y\le 2\}$, $R_1=\{(x,y):0\le x\le 2, 0\le y\le 1\}$, and $R_2=\{(x,y):0\le x\le 2, 1\le y\le 2\}$. Functions $f(x,y)$ and $g(x,y)$ defined on R satisfies

$$\iint_R f(x,y)\,\mathrm{d}A=2,\ \iint_R g(x,y)\,\mathrm{d}A=7,\ \text{and}\ \iint_{R_1} g(x,y)\,\mathrm{d}A=2.$$

(1) $\iint_R 3f(x,y)\,\mathrm{d}A$

Solution

$$\iint_R 3f(x,y)\,\mathrm{d}A=\iint_R 3f(x,y)\,\mathrm{d}A=3\times 2=6$$

(2) $\iint_{R_2} g(x,y)\,\mathrm{d}A$

Solution

$$\iint_{R_2} g(x,y)\,\mathrm{d}A=\iint_R g(x,y)\,\mathrm{d}A-\iint_{R_1} g(x,y)\,\mathrm{d}A=7-2=5.$$

(3) $\iint_R [2g(x,y)+3f(x,y)]\,\mathrm{d}A$

Solution

$$\iint_R [2g(x,y)+3f(x,y)]\,\mathrm{d}A$$
$$=2\iint_R g(x,y)\,\mathrm{d}A+\iint_R 3f(x,y)\,\mathrm{d}A$$
$$=2\times 7+3\times 2=20.$$

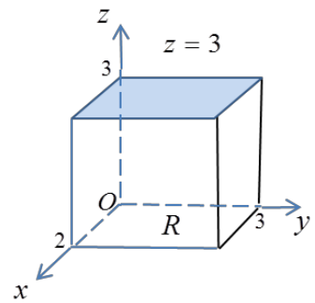
(4) $\iint_{R_2} 6\,\mathrm{d}A$.

Solution

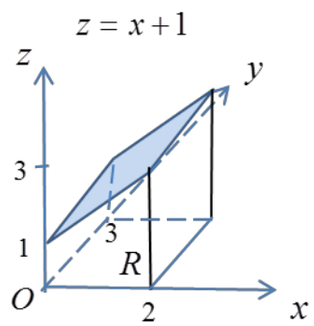
$$\iint_{R_2} 6\,\mathrm{d}A=6\iint_{R_2}\mathrm{d}A=6\times 2=12.$$

2. Sketch the solid whose volume is given by the following double integrals over the rectangle $R=\{(x,y):\ 0\le x\le 2,\ 0\le y\le 3\}$.

(1) $\iint_R 3\,\mathrm{d}A$



(2) $\iint_R (x+1)\,\mathrm{d}A$



Exercise 13-2 Iterated integrals

1. Evaluate the iterated integral $\int_1^2\int_1^3 xy\,\mathrm{d}x\,\mathrm{d}y$.

Solution

$$\int_1^2\int_1^3 xy\,\mathrm{d}x\,\mathrm{d}y=\int_1^2\left[\frac{1}{2}yx^2\right]_{x=1}^{x=3}\mathrm{d}y$$
$$=\frac{1}{2}\int_1^2(9y-y)\,\mathrm{d}y=4\int_1^2 y\,\mathrm{d}y=\left[2y^2\right]_{y=1}^{y=2}=8-2=6.$$

2. Evaluate the indicated double integral over *R*.

$\iint_R xy^3\,\mathrm{d}A;\ R=\{(x,y):0\le x\le 1,-1\le y\le 1\}$.

Solution

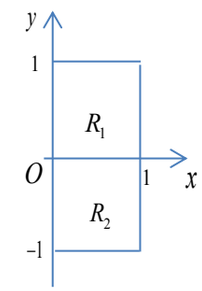
Method 1:

$$\iint_R xy^3\,\mathrm{d}A=\int_0^1\int_{-1}^1 xy^3\mathrm{d}y\mathrm{d}x=\int_0^1\frac{1}{4}xy^4\,|_{y=-1}^{y=1}\,dx=0.$$

Method 2:

Since $f(x,y)=xy^3$ is an odd function in y and the integral region is symmetric w.r.t x-axis, therefore,

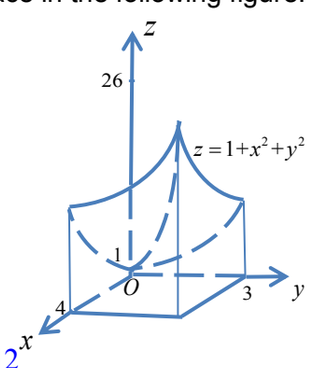
$$\iint_R xy^3\,\mathrm{d}A=0$$



3. Find the volume under the surface in the following figure.

Solution

$$\iint_R\left(1+x^2+y^2\right)\mathrm{d}A$$
$$=\int_0^4\int_0^3\left(1+x^2+y^2\right)\mathrm{d}y\mathrm{d}x$$
$$=\int_0^4\left(y+x^2y+\frac{1}{3}y^3\right)|_{y=0}^{y=3}\,dx$$
$$=\int_0^4(12+3x^2)dx=(12x+x^3)\,|_0^4=112$$



Exercises 13-3 Double integrals over nonrectangular regions

1. Evaluate the iterated integrals.

(1) $\int_1^2\int_0^{x-1} x\,\mathrm{d}y\,\mathrm{d}x$

Solution:

$$\int_1^2\int_0^{x-1} x\mathrm{d}y\mathrm{d}x=\int_1^2[xy]_{y=0}^{y=x-1}dx=\int_1^2 x^2-xdx=\left[\frac{1}{3}x^3-\frac{1}{2}x^2\right]_1^2=\frac{5}{6}.$$

(2) $\int_0^2\int_0^{\sqrt{4-x^2}}(x+y)\,\mathrm{d}y\,\mathrm{d}x$

Solution

$$\int_0^2\int_0^{\sqrt{4-x^2}}(x+y)\,\mathrm{d}y\,\mathrm{d}x$$

$$=\int_0^2(xy+\frac{1}{2}y^2)\,|_{y=0}^{y=\sqrt{4-x^2}}\,\mathrm{d}x=\int_0^2(x\sqrt{4-x^2}+2-\frac{1}{2}x^2)\,\mathrm{d}x$$

$$=-\frac{1}{2}\int_0^2(4-x^2)^{\frac{1}{2}}\,\mathrm{d}(4-x^2)+\int_0^2(2-\frac{1}{2}x^2)\,\mathrm{d}x$$

$$=-\frac{1}{2}\cdot\frac{1}{\frac{1}{2}+1}(4-x^2)^{\frac{3}{2}}\Big|_0^2+(2x-\frac{1}{6}x^3)\,|_0^2=\frac{8}{3}+4-\frac{4}{3}=\frac{16}{3}$$

2. Evaluate the given double integral by changing it to an iterated integral.

(1) $\iint_S xy\,\mathrm{d}A$; S is the region bounded by $y=x^2$ and $y=1$.

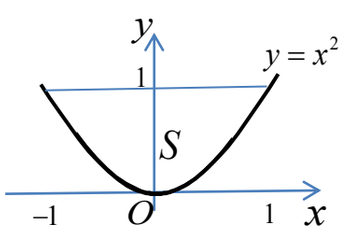
Solution: Method 1:

$$\iint_S xy\,\mathrm{d}A=\int_{-1}^1\int_{x^2}^1 xy\,\mathrm{d}y\mathrm{d}x=\int_{-1}^1\frac{1}{2}xy^2|_{y=x^2}^1\mathrm{d}x=\frac{1}{2}\int_0^1(x-x^5)\mathrm{d}x=0$$

Method 2:

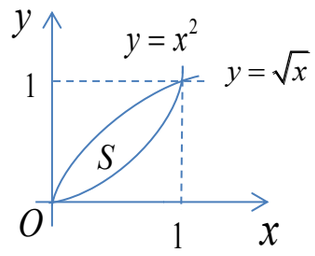
$$\iint_S xy\,\mathrm{d}A=\int_0^1\int_{-\sqrt{y}}^{\sqrt{y}}xy\,\mathrm{d}x\mathrm{d}y$$

$$=\int_0^1\frac{1}{2}yx^2|_{x=-\sqrt{y}}^{\sqrt{y}}\,\mathrm{d}y=\int_0^1 0\,\mathrm{d}y=0$$



Method 3: Since $f(x,y)=xy$ is an odd function in x, and S is symmetric with respect to y-axis, we can conclude that the integral is 0.

(2) $\iint_S(x^2+2y)\,\mathrm{d}A$; S is the region between $y=x^2$ and $y=\sqrt{x}$.



Solution

$$\iint_S(x^2+2y)\,\mathrm{d}A$$

$$=\int_0^1\int_{x^2}^{\sqrt{x}}(x^2+2y)\mathrm{d}y\mathrm{d}x$$

$$=\int_0^1(x^2y+y^2)\,|_{y=x^2}^{\sqrt{x}}\mathrm{d}x$$

$$=\int_0^1(x^{\frac{5}{2}}+x-2x^4)\mathrm{d}x$$

$$=(\frac{2}{7}x^{\frac{7}{2}}+\frac{1}{2}x^2-\frac{2}{5}x^5)\,|_0^1=\frac{27}{70}$$

(3) $\iint_S\frac{2}{1+x^2}\,\mathrm{d}A$; S is the triangular region with vertices at $(0,0),(2,2)$, and $(0,2)$.

Solution

$$\iint_S\frac{2}{1+x^2}\,\mathrm{d}A=\int_0^2\int_x^2\frac{2}{1+x^2}\mathrm{d}y\mathrm{d}x$$

$$=\int_0^2\frac{2}{1+x^2}y\,|_x^2\,\mathrm{d}x$$

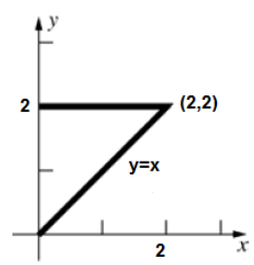
$$=\int_0^2\frac{2}{1+x^2}(2-x)\mathrm{d}x$$

$$=\int_0^2\frac{4}{1+x^2}\mathrm{d}x-\int_0^2\frac{2x}{1+x^2}\mathrm{d}x$$

$$=4\arctan x\,|_0^2-\int_0^2\frac{\mathrm{d}(x^2+1)}{1+x^2}$$

$$=4\arctan 2-\ln(1+x^2)\,|_0^2$$

$$=4\arctan 2-\ln 5$$



3. Find the volume of the solid in the first octant bounded by the cylinder $y=x^2$ and the planes $x=0$, $z=0$, and $y+z=1$ by an iterated integration.

Solution

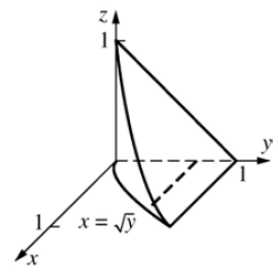
$$z=1-y$$

$$V=\iint_S(1-y)dA$$

$$=\int_0^1\int_0^{\sqrt{y}}(1-y)\mathrm{d}x\mathrm{d}y$$

$$=\int_0^1(x-xy)\,|_{x=0}^{x=\sqrt{y}}\mathrm{d}y=\int_0^1[\sqrt{y}-y\sqrt{y}]\mathrm{d}y$$

$$=\left(\frac{2}{3}y^{\frac{3}{2}}-\frac{2}{5}y^{\frac{5}{2}}\right)\Big|_0^1=\frac{2}{3}-\frac{2}{5}=\frac{4}{15}$$



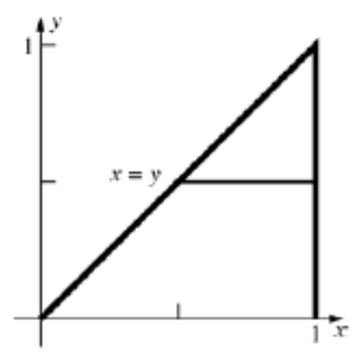
4. Write the given iterated integral as an iterated integral with the order of integration interchanged. *Hint:* Begin by sketching a region S and representing it in two ways, as in Example 5.

(1) $\int_0^1\int_0^x f(x,y)\,\mathrm{d}y\,\mathrm{d}x$

Solution

$$\int_0^1\int_0^x f(x,y)\,dydx$$

$$=\int_0^1\int_y^1 f(x,y)\,dxdy$$

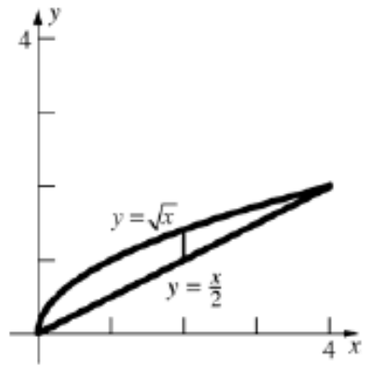


(2) $\int_0^2 \int_{y^2}^{2y} f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

Solution

$\int_0^2 \int_{y^2}^{2y} f(x,y)\,dxdy$

$= \int_0^4 \int_{\frac{1}{2}x}^{\sqrt{x}} f(x,y)\,dydx$

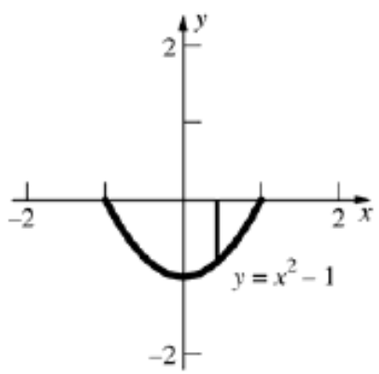


(3) $\int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

Solution

$\int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x,y)dxdy$

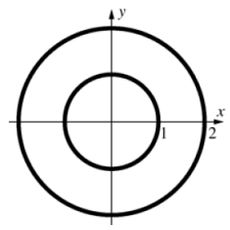
$= \int_{-1}^1 \int_{x^2-1}^0 f(x,y)\,dydx$



5. Evaluate $\iint_S \sin(xy^2)\,\mathrm{d}A$ where S is the annulus $\{(x,y): 1 \le x^2+y^2 \le 4\}$. *Hint:* Done without thinking this problem is hard; using symmetry, it is trivial.

Solution

$z = f(x,y) = \sin(xy^2)$ is an odd function in x and the integral region is symmetric w.r.t. y-axis, therefore $\iint_S \sin(xy^2)dA = 0$



6. Evaluate $\iint_S \sin(y^3)\,\mathrm{d}A$ where S is the region bounded by $y=\sqrt{x}$, $y=2$, and $x=0$. *Hint:* If one order of integration does not work, try the other.

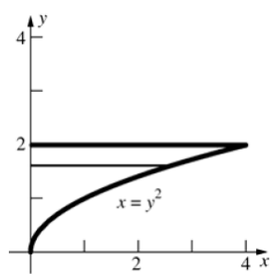
Solution

$\iint_S \sin(y^3)dA = \int_0^2 \int_0^{y^2} \sin(y^3)dxdy$

$= \int_0^2 y^2 \sin(y^3)\,dy = \frac{1}{3}\int_0^2 \sin(y^3)\,d(y^3)$

$= -\frac{1}{3}\cos(y^3)\,|_0^2 = -\frac{1}{3}(\cos 8 - 1)$

$= \frac{1-\cos 8}{3}$



Exercises 13-4 Double integrals in polar coordinates

1. Evaluate the iterated integrals.

(1) $\int_0^{\pi}\int_0^{\sin\theta} r^2\,\mathrm{d}r\,\mathrm{d}\theta$

Solution

$$\int_0^{\pi}\int_0^{\sin\theta} r^2\,dr d\theta=\int_0^{\pi}\frac{1}{3}r^3\,|_{r=0}^{\sin\theta}\,d\theta=\frac{1}{3}\int_0^{\pi}\sin^3\theta\,d\theta$$

$$=\frac{1}{3}\int_0^{\pi}\sin^2\theta\cdot\sin\theta\,d\theta=-\frac{1}{3}\int_0^{\pi}(1-\cos^2\theta)\,d(\cos\theta)$$

$$=\frac{1}{3}(\frac{1}{3}\cos^3\theta-\cos\theta)\,|_0^{\pi}=\frac{4}{9}$$

(2) $\int_0^{\pi}\int_0^{1-\cos\theta} r\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta$

Solution

$$\int_0^{\pi}\int_0^{1-\cos\theta} r\sin\theta\,dr d\theta=\int_0^{\pi}\sin\theta\int_0^{1-\cos\theta} r\,dr d\theta$$

$$=\int_0^{\pi}\sin\theta\left(\frac{1}{2}r^2\,|_0^{1-\cos\theta}\right)d\theta=-\int_0^{\pi}\frac{1}{2}(1-\cos\theta)^2\,d(\cos\theta)$$

$$=\int_0^{\pi}\frac{1}{2}(1-\cos\theta)^2\,d(1-\cos\theta)=\frac{1}{2}\cdot\frac{1}{3}\cdot(1-\cos\theta)^3\,|_0^{\pi}$$

$$=\frac{4}{3}$$

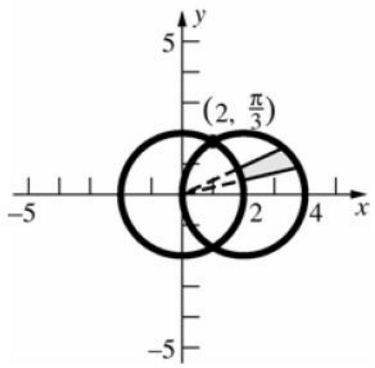
2. Find the area of the given region S by calculating $\iint\limits_S r\,\mathrm{d}r\,\mathrm{d}\theta$. Be sure to make a sketch of the region first.

(1) S is the region inside the circle $r=4\cos\theta$ and outside the circle $r=2$.

Solution

Solve $\begin{cases} r=2 \\ r=4\cos\theta \end{cases}$, we have

$\begin{cases} r=2 \\ \theta=\frac{\pi}{3},-\frac{\pi}{3} \end{cases}$, therefore



$$S=\left\{(r,\theta)\,|\,2\le r\le 4\cos\theta,-\frac{\pi}{3}\le\theta\le\frac{\pi}{3}\right\}$$

$$\iint\limits_S r\,dr d\theta=\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}}\int_2^{4\cos\theta} r\,dr d\theta=\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}}\frac{1}{2}r^2\,|_{r=2}^{4\cos\theta}\,d\theta$$

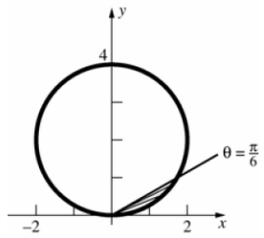
$$=\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}}(8\cos^2\theta-2)\,d\theta=2\int_0^{\frac{\pi}{3}}(8\cdot\frac{1+\cos 2\theta}{2}-2)\,d\theta$$

$$=\int_0^{\frac{\pi}{3}}(4\cos 2\theta+2)\,d(2\theta)=(4\sin 2\theta+4\theta)\,|_0^{\frac{\pi}{3}}=2\sqrt{3}+\frac{4\pi}{3}$$

(2) S is the smaller region bounded by $\theta=\pi/6$ and $r=4\sin\theta$.

Solution

$$S=\left\{(r,\theta)\,|\,0\le r\le 4\sin\theta,0\le\theta\le\frac{\pi}{6}\right\}$$



$$\iint\limits_S r\,dr d\theta=\int_0^{\frac{\pi}{6}}\int_0^{4\sin\theta} r\,dr d\theta$$

$$=\int_0^{\frac{\pi}{6}}\frac{1}{2}r^2\,|_0^{4\sin\theta}\,d\theta=\int_0^{\frac{\pi}{6}}8\sin^2\theta\,d\theta=\int_0^{\frac{\pi}{6}}4(1-\cos 2\theta)\,d\theta$$

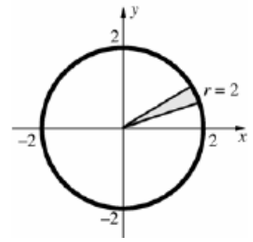
$$=(4\theta-2\sin 2\theta)\,|_0^{\frac{\pi}{6}}=\frac{2}{3}\pi-\sqrt{3}$$

3. Evaluate by using polar coordinates. Sketch the region of integration first.

(1) $\iint\limits_S e^{x^2+y^2}\,\mathrm{d}A$, where S is the region enclosed by $x^2+y^2=4$.

Solution

$$S=\{(r,\theta)\,|\,0\le r\le 2,0\le\theta\le 2\pi\}$$



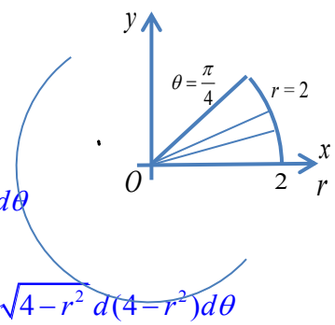
$$\iint\limits_S e^{x^2+y^2}\,dA=\iint\limits_S e^{r^2}\,r\,dr d\theta=\int_0^{2\pi}\int_0^2 e^{r^2}\,r\,dr d\theta$$

$$=\int_0^{2\pi}d\theta\int_0^2 e^{r^2}\,r\,dr=2\pi\cdot(\frac{1}{2}e^{r^2})\,|_0^2=\pi(e^4-1)$$

(2) $\iint\limits_S \sqrt{4-x^2-y^2}\,\mathrm{d}A$, where S is the first quadrant sector of the circle $x^2+y^2=4$ between $y=0$ and $y=x$.

Solution

$$S=\left\{(r,\theta)\,|\,0\le r\le 2,0\le\theta\le\frac{\pi}{4}\right\}$$



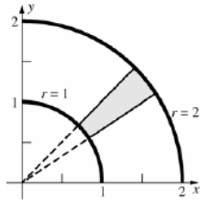
$$\iint\limits_S \sqrt{4-x^2-y^2}\,dA=\iint\limits_S \sqrt{4-r^2}\,r dr d\theta$$

$$=\int_0^{\frac{\pi}{4}}\int_0^2\sqrt{4-r^2}\,r dr d\theta=\int_0^{\frac{\pi}{4}}\int_0^2(-\frac{1}{2})\sqrt{4-r^2}\,d(4-r^2)d\theta$$

$$=\int_0^{\frac{\pi}{4}}(-\frac{1}{2})\cdot\frac{2}{3}\cdot(4-r^2)^{\frac{3}{2}}\,|_{r=0}^2\,d\theta=\int_0^{\frac{\pi}{4}}\frac{8}{3}\,d\theta=\frac{8}{3}\cdot\frac{\pi}{4}=\frac{2\pi}{3}$$

(3) $\iint\limits_S y\,\mathrm{d}A$, where S is the first quadrant polar rectangle inside $x^2+y^2=4$ and outside $x^2+y^2=1$.

Solution



$$S=\left\{(r,\theta)\,|\,1\le r\le 2, 0\le\theta\le\frac{\pi}{2}\right\}$$

$$\iint\limits_S y\,dA=\iint\limits_S r\sin\theta\cdot r\,dr\,d\theta$$

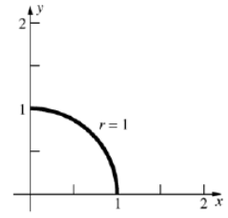
$$=\int_0^{\frac{\pi}{2}}\sin\theta\int_1^2 r^2 dr\,d\theta=\int_0^{\frac{\pi}{2}}\sin\theta\left(\frac{1}{3}r^3\,\Big|_{r=1}^{2}\right)d\theta$$

$$=\frac{7}{3}\int_0^{\frac{\pi}{2}}\sin\theta\,d\theta$$

$$=-\frac{7}{3}\cos\theta\,\Big|_0^{\frac{\pi}{2}}=\frac{7}{3}$$

(4) $\int_0^1\int_0^{\sqrt{1-y^2}}\sin(x^2+y^2)\,\mathrm{d}x\,\mathrm{d}y$

Solution



$$S=\left\{(r,\theta)\,|\,0\le r\le 1, 0\le\theta\le\frac{\pi}{2}\right\}$$

$$\int_0^1\int_0^{\sqrt{1-y^2}}\sin(x^2+y^2)\,dxdy$$

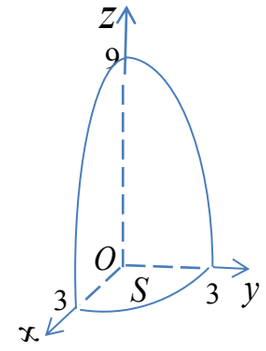
$$=\iint\limits_S\sin(r^2)\cdot r\,dr\,d\theta=\int_0^{\frac{\pi}{2}}\int_0^1\sin(r^2)\cdot r\,dr\,d\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}d\theta\int_0^1\sin(r^2)\,d(r^2)=-\frac{1}{2}\int_0^{\frac{\pi}{2}}\left(\cos r^2\,\Big|_0^1\right)d\theta$$

$$=\int_0^{\frac{\pi}{2}}\frac{1-\cos 1}{2}\,d\theta=\frac{(1-\cos 1)\pi}{4}$$

4. Find the volume of the solid in the first octant bounded by the surface $z=9-x^2-y^2$ and the coordinate planes by an iterated integration.

Solution



$$S=\left\{(r,\boldsymbol{\theta})\,|\,0\le r\le 3, 0\le\boldsymbol{\theta}\le\frac{\boldsymbol{\pi}}{2}\right\},$$

$$z=f(r\cos\boldsymbol{\theta}, r\sin\boldsymbol{\theta})=9-r^2$$

$$V=\iint\limits_S(9-r^2)\,\mathrm{d}A$$

$$=\int_0^{\frac{\pi}{2}}\int_0^3(9-r^2)\,r\,\mathrm{d}r\,\mathrm{d}\theta$$

$$=\int_0^{\frac{\pi}{2}}\left(\frac{9r^2}{2}-\frac{r^4}{4}\right)\Bigg|_{r=0}^{3}\mathrm{d}x$$

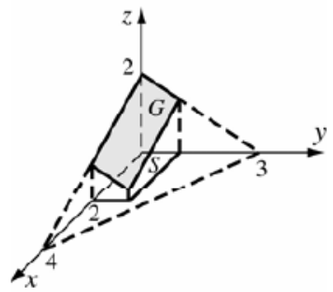
$$=\frac{81}{8}\pi$$

Exercises 13-6 Surface Area

In Problems 1-4, find the area of the indicated surface. Make a sketch in each case.

1. The part of the plane $3x+4y+6z=12$ that is above the rectangle in the *xy*-plane with vertices $(0,0),(2,0),(2,1)$ and $(0,1)$.

Solution:



$S=\{(x,y)|0\le x\le 2, 0\le y\le 1\}$

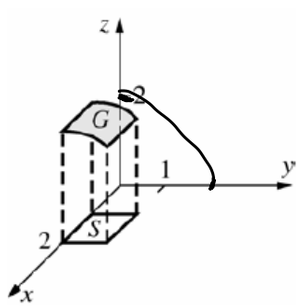
$z=f(x,y)=2-\frac{1}{2}x-\frac{2}{3}y,\ \ z_x=-\frac{1}{2}, z_y=-\frac{2}{3}$

$A(G)=\iint_S\sqrt{1+(z_x)^2+(z_y)^2}\,dA=\iint_S\frac{\sqrt{61}}{6}dA$

$=\int_0^2\int_0^1\frac{\sqrt{61}}{6}dydx=\frac{\sqrt{61}}{3}$

2. The part of the surface $z=\sqrt{4-y^2}$ that is directly above the square in the *xy*-plane with vertices $(1,0),(2,0),(2,1)$ and $(1,1)$.

Solution:



$S=\{(x,y)|0\le x\le 2, 0\le y\le 1\}$

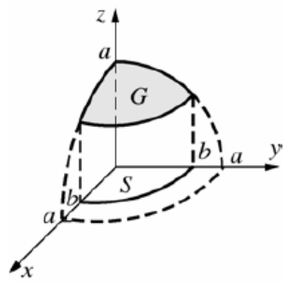
$z_x=0,\ \ z_y=-\frac{y}{\sqrt{4-y^2}}$

$A(G)=\iint_S\sqrt{1+(z_x)^2+(z_y)^2}\,dA=\iint_S\sqrt{1+\frac{y^2}{4-y^2}}\,dA$

$=\int_1^2\int_0^1\frac{2}{\sqrt{4-y^2}}dydx=2\int_1^2dx\int_0^1\frac{1}{\sqrt{1-\left(\frac{y}{2}\right)^2}}d(\frac{y}{2})$

$=2\int_0^1\left[\arcsin\frac{y}{2}\right]_0^1dx=2\arcsin\frac{1}{2}=\frac{\pi}{3}.$

3. The part of the sphere $x^2+y^2+z^2=a^2$ inside the circular cylinder $x^2+y^2=b^2$, where $0<b\le a$.



Solution:

The figure shows the part of the indicated surface that lies in the first octant:

$S=\{(x,y)|x^2+y^2\le b^2, 0\le y\le 1\}=\left\{(r,\theta)\,|0\le r\le b,\ 0\le\theta\le\frac{\pi}{2}\right\}$

$z=\sqrt{a^2-x^2-y^2},\ \ z_x=\frac{x}{\sqrt{a^2-x^2-y^2}},\ \ z_y=\frac{y}{\sqrt{a^2-x^2-y^2}}.$

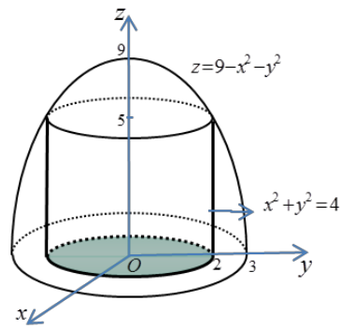
$A(G)=8\iint_S\sqrt{1+(z_x)^2+(z_y)^2}\,dA=8\iint_S\frac{a}{\sqrt{a^2-x^2-y^2}}dA$

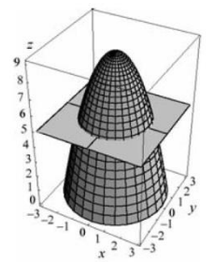
$=8\int_0^{\frac{\pi}{2}}\int_0^b\frac{a}{\sqrt{a^2-r^2}}rdrd\theta=8a\cdot\frac{\pi}{2}\int_0^b\frac{1}{\sqrt{a^2-r^2}}rdr$

$=4a\pi\left[-\sqrt{a^2-r^2}\right]_0^b=4a\pi\left(a-\sqrt{a^2-b^2}\right).$

4. The part of $z=9-x^2-y^2$ above the plane $z=5$.

Solution:





$\begin{cases}z=9-x^2-y^2\\z=5\end{cases}\Rightarrow\begin{cases}x^2+y^2=4\\z=5\end{cases}$

$S=\{(x,y)|x^2+y^2\le 4\}=\{(r,\theta)|0\le r\le 2, 0\le\theta\le 2\pi\}$

$z_x=-2x,\ \ z_y=-2y$

$A(G)=\iint_S\sqrt{1+(z_x)^2+(z_y)^2}\,dA=\iint_S\sqrt{1+4(x^2+y^2)}\,dA$

$=\iint_S\sqrt{1+4r^2}\ rdrd\theta=\int_0^{2\pi}\int_0^2\sqrt{1+4r^2}\ rdrd\theta$

$=\int_0^{2\pi}\left[\frac{1}{12}\left(1+4r^2\right)^{\frac{3}{2}}\right]_{r=0}^2d\theta=\frac{\left(17^{3/2}-1\right)\pi}{6}$